

I never received not gave help regarding this exam

이과생 ①

$$1. R = (q^4 - q^T q)I + 2q q^T + 2q_4 S(q)$$

$$q = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{bmatrix}, \quad q_4 = \frac{\sqrt{2}}{2}$$

$$q^4 = \frac{1}{2}, \quad q^T q = \frac{1}{2}, \quad q q^T = \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{bmatrix}$$

$$2q_4 S(q) = \sqrt{2} \begin{bmatrix} 0 & -\frac{\sqrt{6}}{6} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} & 0 & -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} & \frac{\sqrt{6}}{6} & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{3} & \frac{\sqrt{2}}{3} \\ \frac{\sqrt{2}}{3} & 0 & -\frac{\sqrt{2}}{3} \\ -\frac{\sqrt{2}}{3} & \frac{\sqrt{2}}{3} & 0 \end{bmatrix}$$

$$\therefore R = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} - \frac{\sqrt{3}}{3} & \frac{1}{3} + \frac{\sqrt{3}}{3} \\ \frac{1}{3} + \frac{\sqrt{3}}{3} & \frac{1}{3} & \frac{1}{3} - \frac{\sqrt{3}}{3} \\ \frac{1}{3} - \frac{\sqrt{3}}{3} & \frac{1}{3} + \frac{\sqrt{3}}{3} & \frac{1}{3} \end{bmatrix}$$

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$$2. \frac{d}{dt} (q_1^2 + q_2^2 + q_3^2 + q_4^2) = 2q_1 \dot{q}_1 + 2q_2 \dot{q}_2 + 2q_3 \dot{q}_3 + 2q_4 \dot{q}_4 \text{ 이며,}$$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} S(\omega) \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \frac{1}{2} q_4 \omega \quad \text{또} \quad \dot{q}_4 = -\frac{1}{2} \omega^T \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \text{ 을 조합하여}$$

나타내면,

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} \quad \text{가 되고, 다시 바꾸어 나타내면}$$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ -q_3 & q_4 & q_1 & q_2 \\ -q_2 & q_1 & q_4 & q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \\ 0 \end{bmatrix} \quad \text{3. 나눌 수 있어,}$$

이제 행렬식을 이용하여 다시 나타내면 다음과 같다.

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \\ 0 \end{bmatrix} = 2 \begin{bmatrix} q_4 & q_3 & -q_2 & -q_1 \\ -q_3 & q_4 & q_1 & -q_2 \\ q_2 & -q_1 & q_4 & -q_3 \\ -q_1 & q_2 & q_3 & q_4 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix}$$

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마지막 행을 따르려면 $2(q_1\dot{q}_1 + q_2\dot{q}_2 + q_3\dot{q}_3 + q_4\dot{q}_4) = 0$

이므로 quaternion이 물체의 kinematics를 만족한다면 이는 $t \geq 0$ 에서 항상 성립하오,

$$2(q_1\dot{q}_1 + q_2\dot{q}_2 + q_3\dot{q}_3 + q_4\dot{q}_4) = \frac{d}{dt} (q_1^2 + q_2^2 + q_3^2 + q_4^2) = 0 \text{ 이}$$

$t \geq 0$ 에서 항상 성립하게 되며, $q_1^2(0) + q_2^2(0) + q_3^2(0) + q_4^2(0) = 1$ 이므로,

$$q_1^2(t) + q_2^2(t) + q_3^2(t) + q_4^2(t) = 1 \text{ 이 } t \geq 0 \text{ 에서 언제나 성립된다.}$$

3. a. initial attitude가 $R_0 = e^{S(\vec{\alpha})}$ 로 표현된다면 $R = e^{-S(\vec{\alpha})\phi}$ 이므로

$\phi_0 = -1$ 라 하오.

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rest-to-rest maneuver를 Impulsive control로 설계한다면,

$$t=0, \quad \phi(0) = \phi_0 = -1, \quad \dot{\phi}(0) = 0$$

$$t=0^+, \quad \phi(0^+) = \phi_0 = -1, \quad \dot{\phi}(0^+) = 0 + \Delta\dot{\phi}(0)$$

$$t \in [0^+, t_f^-], \quad \dot{\phi}(t) = \Delta\dot{\phi}(0), \quad \phi(t) = \Delta\dot{\phi}(0)t + \phi_0$$

$$t=t_f, \quad \phi(t_f) = 0 = \Delta\dot{\phi}(0)t_f + \phi_0 \quad (1)$$

$$\dot{\phi}(t_f) = 0 = \Delta\dot{\phi}(0) + \Delta\dot{\phi}(t_f^-) \quad (2)$$

위의 과정을 설계할 수 있어,

$\phi_0 = -1$ 이고 ①에서 $-\Delta\dot{\phi}(0) = -\phi_0/t_f$ 이며,

$\Delta\dot{\phi}(0)$ 은 $\pm 0.01 \text{ rad/sec}$ 이 될수 있으므로 t_f 가 양수가 되도록

$+0.01$ 로 선택하면, $t_f = 100 \text{ sec}$ 이되고,

②에서 $-\Delta\dot{\phi}(0) = \Delta\dot{\phi}(t_f^-)$ 가 성립하므로,

$\Delta\dot{\phi}(t_f^-) = -0.01 \text{ rad/sec}$ 이된다.

따라서 $\dot{\theta}(t)$ 는 $\dot{\theta}(t) = \Delta\dot{\phi}(0) = 0.01 \text{ rad/sec}$ 이고,

이를 이용하여 $\vec{\omega}$ 는 body-fixed frame으로 나타내면,

$\vec{\omega} = \vec{\alpha} \dot{\phi}(t) = \begin{bmatrix} 0.424 \\ -0.566 \\ 0.101 \end{bmatrix} \times 10^{-2} \text{ rad/sec}$ 이된다. X

b. control moment vector 는,

$M = I \dot{\vec{\omega}}$ 이고, ①에서 설계한 것은 two-impulsive control 이므로,

t_0 일때 $\ddot{\phi}(t_0) = +0.01 \text{ rad/sec}^2$, t_f 일때 $\ddot{\phi}(t_f) = -0.01 \text{ rad/sec}^2$ 이고,

이때 $\dot{\vec{\omega}}(t) = \vec{\alpha} \ddot{\phi}(t) = \begin{bmatrix} 0.00424 \\ -0.00566 \\ 0.00101 \end{bmatrix} \text{ rad/sec}^2$ X

$\dot{\vec{\omega}}(t_f) = \vec{\alpha} \ddot{\phi}(t_f) = \begin{bmatrix} -0.00424 \\ 0.00566 \\ -0.00101 \end{bmatrix} \text{ rad/sec}^2$ 이된다. 따라서,

$\vec{M}(t) = \begin{cases} t=t_0=0, & \vec{M} = \begin{bmatrix} 0.848 \\ -1.132 \\ 1.414 \end{bmatrix} \text{ kg} \cdot \text{m}^2 \cdot \text{rad/sec}^2 \\ t_0 < t < t_f, & \vec{M} = 0 \\ t=t_f=100 \text{ sec} & \rightarrow \vec{M} = \begin{bmatrix} -0.848 \\ 1.132 \\ -1.414 \end{bmatrix} \text{ kg} \cdot \text{m}^2 \cdot \text{rad/sec}^2 \end{cases}$

④ 이과광원

C. $a\pi$ 까지 구한대요, $\Delta\phi(0) = -\phi_0/t_f = 0.01 \text{ rad/sec}$ 이고요,
 $t_f = 100 \text{ sec}$ 이된다. X

4. $I_y \Delta\ddot{\theta} = 3\omega_0^2 (-I_x + I_z) \Delta\theta + M_y$

$$\rightarrow \underbrace{\begin{bmatrix} \Delta\dot{\theta} \\ \Delta\ddot{\theta} \end{bmatrix}}_{\dot{X}} = \underbrace{\begin{bmatrix} 0 & 1 \\ \frac{3\omega_0^2(-I_x+I_z)}{I_y} & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} \Delta\theta \\ \Delta\dot{\theta} \end{bmatrix}}_X + \begin{bmatrix} 0 \\ \frac{M_y}{I_y} \end{bmatrix}$$

A를 이용하여 state-transition matrix를 구해보면,

$$\Phi(t) = \mathcal{L}^{-1}((sI-A)^{-1}) = \mathcal{L}^{-1}\left(\begin{bmatrix} s & -1 \\ 1.08 \times 10^{-6} & s \end{bmatrix}^{-1}\right)$$

($1.08 \times 10^{-6} = a^2$ 이라 하면,)

$$= \mathcal{L}^{-1}\left(\frac{1}{s^2 + a^2} \begin{bmatrix} s & +1 \\ -a^2 & s \end{bmatrix}\right)$$

$$= \begin{bmatrix} \cos at & \frac{1}{a} \sin at \\ -a \sin at & \cos at \end{bmatrix}$$

이제 이/를 다시 나타내보면,
 /1625.1 m/s

⑤ 이과항.

$$\begin{bmatrix} \Delta\theta(t) \\ \Delta\dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos at & \frac{1}{a} \sin at \\ -a \sin at & \cos at \end{bmatrix} \begin{bmatrix} \Delta\theta_0 \\ \Delta\dot{\theta}_0 \end{bmatrix}$$

이와 같이 되며,

time 이 지라되지 않을 경우

$$\Delta\theta(t) = (\cos at) \Delta\theta_0 + \frac{1}{a} \sin at (\Delta\dot{\theta}_0) = 0$$

$$\Delta\dot{\theta}(t) = (-a \sin at) \Delta\theta_0 + \cos at (\Delta\dot{\theta}_0) = 0$$

이 2개의 eqn을 만족하는 두개의 변수 $t, \Delta\theta_0$ 를 찾는다.
single impulse 만으로도 control이 가능하게 된다.

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5. a.
$$\begin{bmatrix} \Delta\ddot{\theta} \\ \Delta\dot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{3\omega_0^2(-I_x + I_z)}{I_y} & 0 \end{bmatrix} \begin{bmatrix} \Delta\theta \\ \Delta\dot{\theta} \end{bmatrix}$$

이때 $\frac{3\omega_0^2(-I_x + I_z)}{I_y} = 2.16 \times 10^{-6}$ 이고,

characteristic eqn은,

y축 기준으로

$$s^2 - 2.16 \times 10^{-6} = 0 \text{ 이 되어, } \sqrt{\text{복소평면의 오른쪽}}$$

평면에서 pole이 생기게 되므로 unstable 이 된다,

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Feedback linearization 기법, pitch angle에 대해 이과장전 ⑥
예. b. $\ddot{\theta} + C_{\theta} \dot{\theta} + k_{\theta} (\theta - \theta_d) = 0$ 이라면,

$$R = \begin{matrix} \dot{\theta}_d = 0, & \theta_d = 0 \text{ 이므로 } C_{\theta}, k_{\theta} \text{ 는 } \end{matrix}$$

characteristic polynomial 을 만족하도록 설계하면,

$$M(s) = C_{\theta} = 0.0112, \quad k_{\theta} = 0.000064 \text{ 가 된다.}$$

$$\text{이때 } C_{\theta} > 0, \quad C_{\theta}^2 - 4k_{\theta} = -1.3056 \times 10^{-8} < 0 \text{ 이므로}$$

asymptotically stable 하게 된다.

$$\#1. \vec{g} = [\sqrt{6}/6, \sqrt{6}/6, \sqrt{6}/6, \sqrt{2}/2]$$

$$\text{Let } \vec{g} = [\sqrt{6}/6, \sqrt{6}/6, \sqrt{6}/6]^T \text{ and } g_4 = \sqrt{2}/2.$$

$$\Rightarrow R = (g_4^2 - \vec{g}^T \vec{g}) \mathbf{I} + 2\vec{g}\vec{g}^T + 2g_4 S(\vec{g})$$

$$= (\frac{1}{2} - \cancel{3 \cdot \frac{1}{6}}) \mathbf{I} + 2 \times \frac{1}{6} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} + \frac{\sqrt{2}}{\sqrt{6}} \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 1-\sqrt{3} & 1+\sqrt{3} \\ 1+\sqrt{3} & 1 & 1-\sqrt{3} \\ 1-\sqrt{3} & 1+\sqrt{3} & 1 \end{bmatrix}$$

$$\#2. \text{Let } \vec{g} = [g_1, g_2, g_3]^T, \text{ and } Q = [g^T, g_4]^T$$

$$\dot{\vec{g}} = \frac{1}{2} S(\omega) \vec{g} + \frac{1}{2} g_4 \omega. \quad \& \quad \dot{g}_4 = -\frac{1}{2} \omega^T \vec{g}. \quad \forall t \geq 0$$

$$\text{Claim: } Q^T Q = 1 \quad \forall \text{ time } t \geq 0$$

$$\dot{Q} = \begin{bmatrix} \frac{1}{2} S(\omega) & \frac{1}{2} \omega \\ -\frac{1}{2} \omega^T & 0 \end{bmatrix} Q.$$

$$\frac{d}{dt}(Q^T Q) = 2Q^T \dot{Q} = 2 \cdot \frac{1}{2} Q^T \begin{bmatrix} S(\omega) & \omega \\ -\omega^T & 0 \end{bmatrix} Q = Q^T \begin{bmatrix} S(\omega) & \omega \\ -\omega^T & 0 \end{bmatrix} Q$$

$$= [g_1 \ g_2 \ g_3 \ g_4] \begin{bmatrix} 0 & -\omega_z & \omega_y & \omega_x \\ \omega_z & 0 & -\omega_x & \omega_y \\ -\omega_y & \omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \end{bmatrix}$$

$$= [g_1 \ g_2 \ g_3 \ g_4] \begin{bmatrix} -\omega_z g_2 + \omega_y g_3 + \omega_x g_4 \\ \omega_z g_1 - \omega_x g_3 + \omega_y g_4 \\ -\omega_y g_1 + \omega_x g_2 + \omega_z g_4 \\ -\omega_x g_1 - \omega_y g_2 - \omega_z g_3 \end{bmatrix}$$

$$= -\omega_z g_1 g_2 + \omega_y g_1 g_3 + \omega_x g_1 g_4 + \omega_z g_1 g_2 - \omega_x g_2 g_3 + \omega_y g_2 g_4 \\ - \omega_y g_1 g_3 + \omega_x g_2 g_3 + \omega_z g_3 g_4 - \omega_x g_1 g_4 - \omega_y g_2 g_4 - \omega_z g_3 g_4 \\ = 0.$$

$$\therefore Q^T Q \text{ is constant for any time } t \geq 0$$

$$\therefore Q^T Q = 1. \quad \forall \text{ time } t \geq 0$$

$$\#3. J = \text{diag}(200, 200, 200) \text{ kgm}^2, \text{ rest to rest maneuver. } \Rightarrow \vec{\omega}(0) = 0 = \vec{\omega}(t_f)$$

$$\vec{a} = [0.424, -0.566, 0.701]^T$$

$$R_0 = \exp[S(\vec{a})] = \mathbf{I} + \sin \phi S(\vec{a}) + S(\vec{a}) (1 - \cos \phi) \\ = \mathbf{I} + \sin |S(\vec{a})| + (\vec{a} \vec{a}^T - \mathbf{I})(1 - \cos |S(\vec{a})|) \quad \rightarrow \mathbf{I} = \mathbf{I}_{3 \times 3} \\ = S(\vec{a}) \sin |S(\vec{a})| + \vec{a} \vec{a}^T (1 - \cos |S(\vec{a})|) + \mathbf{I} \cos |S(\vec{a})|.$$

$$R_f = \mathbf{I}_{3 \times 3}. \Rightarrow \phi_0 = 1 \quad \& \quad \phi_f = 0.$$

$$a. R = \exp(S(\vec{a}) \phi(t))$$

$$\Rightarrow \dot{R} = S(\vec{a}) \dot{\phi}(t) \exp(S(\vec{a}) \phi(t)) = S(\vec{a}) \dot{\phi}(t) R.$$

$$\text{Since } \dot{R} = S(\vec{\omega}) R \text{ from rotation matrix kinematics,}$$

$$\Rightarrow S(\vec{\omega}) R = S(\vec{a}) \dot{\phi}(t) R$$

$$\Rightarrow \vec{\omega} = \vec{a} \dot{\phi}(t) \dots (B)$$

Assume that $\omega(t) = \begin{cases} -0.01 \text{ rad/s}^2 & \text{for } 0 \leq t \leq t_s \\ +0.01 \text{ rad/s}^2 & \text{for } t_s < t \leq t_f \end{cases}$

From Euler rotational EoM,

$$\begin{cases} I_x \dot{\omega}_x = (I_y - I_z) \omega_y \omega_z + M_x = M_x \\ I_y \dot{\omega}_y = (I_z - I_x) \omega_x \omega_z + M_y = M_y \\ I_z \dot{\omega}_z = (I_x - I_y) \omega_x \omega_y + M_z = M_z \end{cases} \Rightarrow M = J \ddot{\omega} = J \ddot{\alpha} \phi = J \ddot{\alpha} u \quad \dots (A)$$

$$\therefore \ddot{\omega}(t) = \ddot{\alpha} u(t)$$

$$\Rightarrow \omega(t) = \int_0^t \ddot{\alpha} u(\tau) d\tau$$

$$\text{If } 0 \leq t < t_s, \omega(t) = \int_0^t \ddot{\alpha} u(\tau) d\tau = \ddot{\alpha} \int_0^t -0.01 d\tau = -0.01 t \ddot{\alpha}$$

$$\text{If } t = t_s, \omega(t_s) = \int_0^{t_s} \ddot{\alpha} u(\tau) d\tau = -\ddot{\alpha} 0.01 t_s$$

$$\begin{aligned} \text{If } t_s < t \leq t_f, \omega(t) &= \int_0^t \ddot{\alpha} u(\tau) d\tau \\ &= \int_0^{t_s} \ddot{\alpha} u(\tau) d\tau + \int_{t_s}^t \ddot{\alpha} u(\tau) d\tau \\ &= -0.01 t_s \ddot{\alpha} + 0.01 (t - t_s) \ddot{\alpha} \\ &= 0.01 (2t_s - t) \ddot{\alpha} \end{aligned}$$

$$\Rightarrow \omega(t) = \begin{cases} -0.01 t \ddot{\alpha} & \text{for } 0 \leq t \leq t_s \text{ (rad/s)} \\ -0.01 (2t_s - t) \ddot{\alpha} & \text{for } t_s \leq t \leq t_f \text{ (rad/s)} \end{cases}$$

b. From (A), $M = J \ddot{\alpha} u = 200 \ddot{\alpha} u$

$$\Rightarrow M(t) = \begin{cases} -2 \ddot{\alpha} \text{ kgm}^2/\text{s}^2 & \text{for } 0 \leq t \leq t_s \\ +2 \ddot{\alpha} \text{ kgm}^2/\text{s}^2 & \text{for } t_s \leq t \leq t_f \end{cases}$$

c. Since the problem is rest to rest problem, $\ddot{\omega}(t_f) = 0$

$$\Rightarrow \ddot{\omega}(t_f) = -0.01 (2t_s - t_f) \ddot{\alpha} = 0$$

$$\Rightarrow 2t_s = t_f \therefore t_s = t_f / 2$$

From (B), $\ddot{\omega} = \ddot{\alpha} \phi(t)$,

$$\ddot{\alpha} \phi(t) = \int_0^t \ddot{\alpha} \phi(\tau) d\tau + \ddot{\alpha} = \int_0^t \ddot{\omega}(\tau) d\tau + \ddot{\alpha}$$

If $0 \leq t < t_s = t_f / 2$,

$$\ddot{\alpha} \phi(t) = \int_0^t -0.01 \tau \ddot{\alpha} d\tau + \ddot{\alpha} = \ddot{\alpha} (-0.005 t^2 + 1)$$

$$\Rightarrow \phi(t) = -0.005 t^2 + 1$$

$$\text{If } t = t_s, \phi(t_s) = -0.005 t_s^2 + 1$$

If $t_s < t \leq t_f$,

$$\begin{aligned} \ddot{\alpha} \phi(t) &= \int_0^{t_s} 0.01 \tau \ddot{\alpha} d\tau + \int_{t_s}^t -0.01 (2t_s - \tau) \ddot{\alpha} d\tau + \ddot{\alpha} \\ &= (0.005 t_s^2 - 0.02 t_s (t - t_s) + 0.005 (t^2 - t_s^2) + 1) \ddot{\alpha} \\ &= (0.01 t_s^2 - 0.02 t_s t + 0.005 t^2 + 1) \ddot{\alpha} \\ \Rightarrow \phi(t) &= 0.01 (t_s - t)^2 - 0.005 t^2 + 1 \end{aligned}$$

$$\text{Since } \phi(t_f) = 0, \phi(t_f) = -0.005 t_f^2 + 0.01 (t_s - t_f)^2 + 1 = 0$$

$$\Rightarrow t_f^2 = +400 \dots ??$$

$$\therefore t_f = 20$$

#4. Pitch axis control model. $J_y \ddot{\theta} = 3\omega_0^2 (-I_x + I_z) \theta + M_y$
 $\omega_0 = 0.0012 \text{ rad/sec}$
 $J = \text{diag}(150, 200, 100) \Rightarrow I_x = 150, I_y = 200, I_z = 100.$

test to test maneuver

$$\Delta\theta_0 = 45^\circ = \pi/4 \text{ rad} \quad \& \quad \Delta\theta_f = 0^\circ = 0 \text{ rad}$$

Let assume that this test to test maneuver can be performed using a single impulse, T .

Let t_s be a time that a single impulse is performed.

$$200 \ddot{\theta} = 3\omega_0^2 (-150 + 100) \theta + M_y$$

$$= -150\omega_0^2 \theta + M_y$$

$$\Rightarrow \ddot{\theta} = -(3/4)\omega_0^2 \theta + M_y/200$$

$$\Rightarrow \begin{bmatrix} \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -(3/4)\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ 1/200 \end{bmatrix} M_y$$

If $t \in [0, t_s]$,

$$\begin{bmatrix} \theta(t) \\ \dot{\theta}(t) \end{bmatrix} = \exp\left(\begin{bmatrix} 0 & 1 \\ -(3/4)\omega_0^2 & 0 \end{bmatrix} t\right) \begin{bmatrix} \pi/4 \\ 0 \end{bmatrix}$$

$$\exp\left(\begin{bmatrix} 0 & 1 \\ -a & 0 \end{bmatrix} t\right) = I + At + \frac{1}{2!} (At)^2 + \dots \text{ where } a = \frac{3}{4}\omega_0^2$$

$$= I + At + \frac{1}{2!} (-at^2)I + \frac{1}{3!} (-at^3)A + \dots$$

$$= I \left(\sum_{i=0}^{\infty} \frac{(-at^2)^i}{(2i)!} \right) + A \left(\sum_{i=1}^{\infty} \frac{(-a)^i t^{2i-1}}{(2i-1)!} \right)$$

$$= \cos(\sqrt{a}t)I + \frac{1}{\sqrt{a}} \sin(\sqrt{a}t)A$$

$$= \begin{bmatrix} \cos\sqrt{a}t & \frac{1}{\sqrt{a}} \sin(\sqrt{a}t) \\ -\sqrt{a} \sin(\sqrt{a}t) & \cos\sqrt{a}t \end{bmatrix}$$

$$\begin{bmatrix} \theta(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos\sqrt{a}t & \sin(\sqrt{a}t)/\sqrt{a} \\ -\sqrt{a} \sin\sqrt{a}t & \cos\sqrt{a}t \end{bmatrix} \begin{bmatrix} \pi/4 \\ 0 \end{bmatrix}$$

Thus at $t = t_s$, $\theta(t_s) = (\pi/4) \cos\sqrt{a}t_s = (\pi/4) \cos\sqrt{a}t_s$
 $\dot{\theta}(t_s) = -\sqrt{a}(\pi/4) \sin\sqrt{a}t_s$

At $t = t_s^+$, $\theta(t_s^+) = \theta(t_s)$ & $\dot{\theta}(t_s^+) = \dot{\theta}(t_s) + T$.

If $t \in (t_s, t_f)$,

$$\begin{bmatrix} \theta(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos\sqrt{a}(t-t_s) & \sin\sqrt{a}(t-t_s)/\sqrt{a} \\ -\sqrt{a} \sin\sqrt{a}(t-t_s) & \cos\sqrt{a}(t-t_s) \end{bmatrix} \begin{bmatrix} (\pi/4) \cos\sqrt{a}t_s \\ -\sqrt{a}(\pi/4) \sin\sqrt{a}t_s + T \end{bmatrix}$$

At $t = t_f$,

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\sqrt{a}(t_f-t_s) & \sin\sqrt{a}(t_f-t_s)/\sqrt{a} \\ -\sqrt{a} \sin\sqrt{a}(t_f-t_s) & \cos\sqrt{a}(t_f-t_s) \end{bmatrix} \begin{bmatrix} (\pi/4) \cos\sqrt{a}t_s \\ -\sqrt{a}(\pi/4) \sin\sqrt{a}t_s + T \end{bmatrix}$$

Since $\cos^2\sqrt{a}(t_f-t_s) + \sin^2\sqrt{a}(t_f-t_s) = 1$, STM is invertible $\forall t_s, t_f$.

It means that there exist only one element in $\ker(\text{STM})$ for any t_s and t_f .
 $= [0, 0]^T$

Thus, $\cos\sqrt{a}t_s = 0$ and $-\sqrt{a}(\pi/4) \sin\sqrt{a}t_s + T = 0$.

If $\sqrt{s} t_s = n\pi + \frac{\pi}{2}$, then $\sin \sqrt{s} t_s = 1$.

$$\Rightarrow T = \sqrt{a}(\pi/4) \approx 8.1621e-04 \text{ rad/sec.}$$

If $\sqrt{s} t_s = n\pi + 3\pi/2$, then $\sin \sqrt{s} t_s = -1$.

$$\Rightarrow T = -\sqrt{a}(\pi/4) \approx -8.1621e-04 \text{ rad/sec}$$

The smallest possible t_s is 1511.4995 sec and $T = 8.1621e-04 \text{ rad/s}$

Thus control moment $M_y = 200T = 0.1632 \text{ kgm}^2/\text{sec}^2$.

Hence, if transfer time t_f is greater than 1511.4995 sec, then rest to rest maneuver can be performed using a single impulse.

$\Rightarrow$ The claim is true if $t_f \geq 1511.4995 \text{ sec}$.

The times instant the impulse should applied are $(2n-1)\pi/(2\sqrt{a})$ for any integer $n \in \mathbb{Z}$, and this times are same as the transfer time.

#5. $I_y \ddot{\Delta\theta} = 3\omega_0^2(-I_x + I_z) \Delta\theta + M_y$.

$$\omega_0 = 0.0012 \text{ rad/s. } J = \text{diag}(150, 100, 200) \text{ kgm}^2.$$

a. $M_y = 0 \Rightarrow 100 \ddot{\Delta\theta} = +150\omega_0^2 \Delta\theta$.

$$\Rightarrow \ddot{\Delta\theta} = + (3/2) \omega_0^2 \Delta\theta$$

$$\Rightarrow \begin{bmatrix} \dot{\Delta\theta}(t) \\ \ddot{\Delta\theta}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ + (3/2) \omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta\theta \\ \dot{\Delta\theta} \end{bmatrix}$$

$$\text{eigenvalues: } \lambda^2 - \frac{3}{2} \omega_0^2 = 0 \Rightarrow \lambda = \pm \sqrt{\frac{3}{2}} \omega_0$$

Since there is positive real eigenvalue, then this system is NOT stable whatever an initial state is.

Hence the pitch equilibrium $(\Delta\theta, \dot{\Delta\theta}) = (0, 0)$ is unstable.

b. Let $M_y = -k_p \Delta\theta - k_d \dot{\Delta\theta}$

$$\Rightarrow 100 \ddot{\Delta\theta} = 150\omega_0^2 \Delta\theta - k_p \Delta\theta - k_d \dot{\Delta\theta}$$

$$\Rightarrow \ddot{\Delta\theta} + 0.01 k_d \dot{\Delta\theta} + (0.01 k_p - 1.5\omega_0^2) \Delta\theta = 0$$

$$\Rightarrow \{s^2 + 0.01 k_d s + (0.01 k_p - 1.5\omega_0^2)\} \Delta\theta_s = 0$$

$$\Rightarrow s^2 + 0.01 k_d s + (0.01 k_p - 1.5\omega_0^2) = s^2 + 0.0112s + 0.000064$$

$$\Rightarrow 0.01 k_d = 0.0112 \quad \& \quad 0.01 k_p - 1.5\omega_0^2 = 0.000064$$

$$\Rightarrow k_d = 1.12 \quad \& \quad k_p = 6.616e-03$$

Thus when $M_y = -6.616e-03 \Delta\theta - 1.12 \dot{\Delta\theta}$, the system is stable.

#1. $\vec{r} = (0.5 + u)\vec{s}_1$, $\vec{w} = \vec{s}_3$

A. Let assume that the spacecraft is in equilibrium, then $\ddot{x} = \ddot{y} = \ddot{z} = 0$.

$$(i) \ddot{x} - 2\ddot{y} - x = -\frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3}$$

- Left side: $\ddot{x} - 2\ddot{y} - x = -x = -u + 0.5$

- Right side: $|r_1| = 0.5 = |r_2|$

$$\Rightarrow -\frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3} = +\frac{(1-u)0.5}{0.5^3} - \frac{0.5u}{0.5^3}$$

$$= 4(1-u-u) = 4(1-2u)$$

Then $-u + 0.5 = 4(1-2u) \Rightarrow u = 0.5$. Since $u \neq 0.5$, then the left side and the right side are different

$$(ii) \ddot{y} + 2\ddot{x} - y = -\frac{(1-u)y}{r_1^3} - \frac{uy}{r_2^3}$$

- Left side: $\ddot{y} + 2\ddot{x} - y = 0$

- Right side: Since $y = 0$, 0.

Thus the left and the right side are same.

$$(iii) \ddot{z} = -\frac{(1-u)z}{r_1^3} - \frac{uz}{r_2^3} \Rightarrow \text{both sides are zero.}$$

Therefore, the spacecraft is not in equilibrium

There is only S_1 -axis acceleration.

$$\Rightarrow \ddot{x} - u + 0.5 = 4(1-2u)$$

$$\Rightarrow \ddot{x} = 7(0.5 - u)$$

Thus, the acceleration $\ddot{\vec{r}} = 7(0.5 - u)\vec{s}_1$.

B. For keeping the spacecraft in equilibrium at the midway point, an acceleration of the spacecraft should be zero.

Since only S_1 -coordinate component is not in equilibrium, it is enough to control only S_1 -coordinate component

$$\ddot{x} - 2\ddot{y} - x = -\frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3} + u$$

$$\Rightarrow -u + 0.5 = 4(1-2u) + u$$

$$\Rightarrow u = 7(u - 0.5)$$

Hence the control acceleration vector is $\vec{U} = 7(u - 0.5)\vec{s}_1$.

#2. Let $R = R_x(\phi)R_y(\theta)R_z(\psi)$

$$\Rightarrow \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \dot{\vec{\phi}} + \dot{\vec{\theta}} + \dot{\vec{\psi}} = R_x(\phi)R_y(\theta) \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + R_x(\phi) \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$

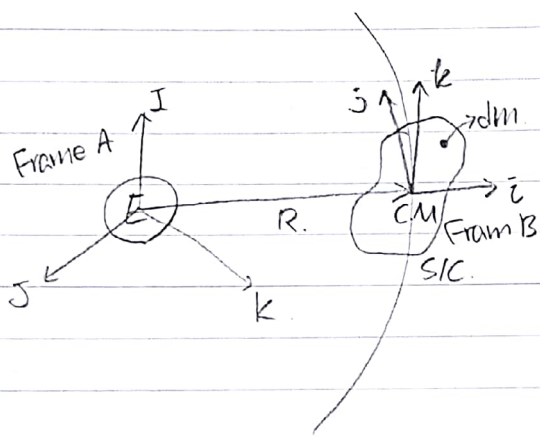
$$= \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ \sin\phi \sin\theta & \cos\phi & \sin\phi \cos\theta \\ \cos\phi \sin\theta & -\sin\phi & \cos\phi \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & \sin\theta \\ 0 & \cos\phi & \sin\phi \cos\theta \\ 0 & -\sin\phi & \cos\phi \cos\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

$$\therefore \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & \sin\theta \\ 0 & \cos\phi & \sin\phi \cos\theta \\ 0 & -\sin\phi & \cos\phi \cos\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

#3

A. $I = \text{diag}(A, B, C)$, $\vec{M} = [M_1, M_2, M_3]^T$, $\vec{\omega} = [\omega_1, \omega_2, \omega_3]^T$



$$H = \int_B \vec{r} \times \vec{v} \, dm = \int_B \vec{r} \times (\vec{\omega} \times \vec{r}) \, dm$$

$$= \int_B (r^T r - r r^T) \vec{\omega} \, dm$$

$$= \int_B (r^T r - r r^T) \, dm \vec{\omega}$$

$$= \int_B \begin{bmatrix} r_y^2 + r_z^2 & -r_x r_y & -r_x r_z \\ -r_x r_y & r_x^2 + r_z^2 & -r_y r_z \\ -r_x r_z & -r_y r_z & r_x^2 + r_y^2 \end{bmatrix} \, dm \vec{\omega}$$

$$= I \vec{\omega}$$

By Coriolis's theorem, $\left(\frac{d}{dt} \vec{H}\right)^A = \left(\frac{d}{dt} \vec{H}\right)^B + \vec{\omega} \times \vec{H}$.

$$\Rightarrow I \dot{\vec{\omega}} = \vec{\omega} \times \vec{H} + \vec{M} = \vec{\omega} \times I \vec{\omega} + \vec{M}$$

$$\therefore I \dot{\vec{\omega}} - \vec{\omega} \times I \vec{\omega} = \vec{M}$$

Since $I = \text{diag}(A, B, C)$,

$$\begin{cases} A \dot{\omega}_1 = (B - C) \omega_2 \omega_3 + M_1 \\ B \dot{\omega}_2 = (C - A) \omega_1 \omega_3 + M_2 \\ C \dot{\omega}_3 = (A - B) \omega_1 \omega_2 + M_3 \end{cases}$$

B. Let $\vec{M} = 0$, then the kinetic energy $T = \frac{1}{2} \vec{\omega}^T I \vec{\omega}$

$$\frac{d}{dt} T = \frac{d}{dt} \left(\frac{1}{2} \vec{\omega}^T I \vec{\omega} \right) = \vec{\omega}^T I \dot{\vec{\omega}} = \vec{\omega}^T (\vec{\omega} \times I \vec{\omega})$$

Since $a \cdot (b \times c) = \det([a, b, c]^T)$,

$$\frac{d}{dt} T = (\vec{\omega}) \cdot (\vec{\omega} \times I \vec{\omega}) = \det([\vec{\omega}, \vec{\omega}, I \vec{\omega}]^T) = 0$$

Thus T is conserved. Let $H = |I \vec{\omega}| \neq 0$.

$$\frac{d}{dt} H^2 = 2 \vec{H} \cdot \dot{\vec{H}} = (I \dot{\vec{\omega}})^T (I \vec{\omega}) = (\vec{\omega} \times I \vec{\omega})^T I \vec{\omega}$$

$$= \det([I \vec{\omega}, \vec{\omega}, I \vec{\omega}]) = 0$$

Thus, H is conserved.

#4. $I_y \ddot{\Delta\theta} = 3\omega_0^2 (-I_x + I_z) \Delta\theta + M_y$

$\omega_0 = 0.0012 \text{ rad/sec}$

$I = \text{diag}(150, 100, 200) \text{ kgm}^2$

A. $100 \ddot{\Delta\theta} = 3\omega_0^2 (-150 + 200) \Delta\theta$

$\Rightarrow \ddot{\Delta\theta} = 0.03\omega_0^2 \cdot 50 \Delta\theta = 1.5\omega_0^2 \Delta\theta$

$\Rightarrow \begin{bmatrix} \ddot{\Delta\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1.5\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta\theta \\ \dot{\Delta\theta} \end{bmatrix}$

$\Rightarrow \text{eigenvalues } \lambda^2 - 1.5\omega_0^2 = 0$

$\Rightarrow \lambda = \pm 1.4697 \times 10^{-3}$

Since there exists positive eigenvalue, then this system is NOT stable.

whatever an initial state is.

B. Let $M_y = -k_p \Delta\theta - k_d \dot{\Delta\theta}$

$\Rightarrow \ddot{\Delta\theta} = 1.5\omega_0^2 \Delta\theta - 0.01k_p \Delta\theta - 0.01k_d \dot{\Delta\theta}$

$\Rightarrow \ddot{\Delta\theta} + 0.01k_d \dot{\Delta\theta} + (0.01k_p - 1.5\omega_0^2) \Delta\theta = 0$

$\Rightarrow (s^2 + 0.01k_d s + (0.01k_p - 1.5\omega_0^2)) \Delta\theta = 0$

$\Rightarrow s^2 + 0.01k_d s + (0.01k_p - 1.5\omega_0^2) = 0$

$\Rightarrow 0.01k_d = 0.0112 \text{ \& } 0.01k_p = 1.5\omega_0^2 + 0.000064$

$\Rightarrow k_d = 1.12 \text{ \& } k_p = 6.616 \times 10^{-3}$

$\therefore M_y = -6.616 \times 10^{-3} \Delta\theta - 1.12 \dot{\Delta\theta}$

#5. Rotational matrix. R

An element of special orthogonal group $SO(3)$

Let $\{i, j, k\}$ be a basis of rotating frame and $\{I, J, K\}$ be a basis of a standard frame, Then R is formed by projection of i, j, k to I, J, K .

$$R = \begin{bmatrix} i \cdot I & i \cdot J & i \cdot K \\ j \cdot I & j \cdot J & j \cdot K \\ k \cdot I & k \cdot J & k \cdot K \end{bmatrix}$$

$\exists$ 9 parameters. $\nexists$ Singularity

- Euler angles

$\exists$ 3 parameters. $\exists$ Singularity at $\theta = \pm \pi/2$

$\exists$ 12 combinations.

Usually use 3-2-1 OR 3-1-3 combination

- Quaternion

$\exists$ 4 parameter $\nexists$ Singularity

Defined using Euler principle axis $\vec{a}$ and angle ϕ

$\vec{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} = \sin \frac{\phi}{2} \vec{a} \quad q_4 = \cos \frac{\phi}{2}$

Convenience for analysis.

#1. $T = 24h = 86400 \text{ s} \Rightarrow \omega_0 = 2\pi/T = 7.2722 \times 10^{-5} \text{ rad/sec}$
 $(x_0, y_0, z_0) = (-110, 70, 10) \text{ km}$ $(\dot{x}_0, \dot{y}_0, \dot{z}_0) = (-10, 20, 1) \text{ m/sec}$
 Two impulsive maneuver.

$t_f = 2h = 7200 \text{ sec}$

$(x_f, y_f, z_f) = 0 \text{ km}$ & $(\dot{x}_f, \dot{y}_f, \dot{z}_f) = 0 \text{ m/sec}$

$\omega t_f = \pi/6 \Rightarrow \sin(\omega t_f) = 1/2$ & $\cos(\omega t_f) = \sqrt{3}/2$

$$\Rightarrow \dot{y}(0^+) = \frac{6x_0(\omega t_f - \sin(\omega t_f)) - y_0 \omega \sin(\omega t_f) - 2\omega x_0(4 - 3\cos(\omega t_f))(1 - \cos(\omega t_f))}{(4\sin(\omega t_f) - 3\omega t_f)\sin\omega t_f + 4(1 - \cos(\omega t_f))^2}$$

$$6x_0(\omega t_f - \sin(\omega t_f)) - y_0 \omega \sin\omega t_f$$

$$= (6(-110)(\pi/6 - 1/2) - 70)7.2722 \times 10^{-5} \cdot \frac{1}{2} \approx -3.1116 \times 10^{-3}$$

$$2\omega x_0(4 - 3\cos\omega t_f)(1 - \cos\omega t_f)$$

$$= 2 \times 7.2722 \times 10^{-5} \times (-110)(4 - 3\sqrt{3}/2)(1 - \sqrt{3}/2) \approx -3.0049 \times 10^{-3}$$

$\Rightarrow \text{Numerator} = -1.0666 \times 10^{-4}$

$$(4\sin(\omega t_f) - 3\omega t_f)\sin\omega t_f + 4(1 - \cos(\omega t_f))^2$$

$$= (4/2 - \pi/2)/2 + 4(1 - \sqrt{3}/2)^2 \approx 0.2864 = \text{Denominator}$$

$$\Rightarrow \dot{y}(0^+) = \frac{-1.0666 \times 10^{-4}}{0.2864} \approx -3.7242 \times 10^{-4}$$

$$\dot{x}(0^+) = -\frac{\omega x_0(4 - 3\cos\omega t_f) + 2(1 - \cos\omega t_f)y_0}{\sin\omega t_f}$$

$$= -2\omega x_0(4 - 3\cos\omega t_f) - 4(1 - \cos\omega t_f)y_0$$

$$= +2 \cdot 110 \cdot 7.2722 \times 10^{-5} (4 - 3\sqrt{3}/2) + 4(1 - \sqrt{3}/2)(3.7242 \times 10^{-4})$$

$$\approx 0.02263$$

$$\dot{z}(0^+) = -z_0 \omega \cot(\omega t_f)$$

$$= -10(7.2722 \times 10^{-5}) \cot(\pi/6) \approx -1.2596 \times 10^{-3}$$

$$\Rightarrow \vec{D}_0 = (\dot{x}(0^+) - \dot{x}_0)\vec{i} + (\dot{y}(0^+) - \dot{y}_0)\vec{j} + (\dot{z}(0^+) - \dot{z}_0)\vec{k}$$

$$= (0.03263\vec{i} - 0.0204\vec{j} - 2.2596 \times 10^{-3}\vec{k}) \text{ km/s}$$

$$\Rightarrow \Delta \vec{D}_0 = 0.03263 \text{ km/s}$$

$$\dot{x}(t) = \dot{x}_0 \cos(\omega t) + (3x_0\omega + 2\dot{y}_0)\sin(\omega t)$$

$$\dot{x}(t_f) = 0.02263 \times \sqrt{3}/2 + (-330 \cdot 7.2722 \times 10^{-5} + 2 \times (-3.7242 \times 10^{-4}))/2$$

$$\approx 7.2266 \times 10^{-3} \text{ km/s} = 7.2266 \text{ m/s}$$

$$\dot{y}(t) = (6x_0\omega + 4\dot{y}_0)\cos(\omega t) - 2\dot{x}_0\sin\omega t - 6\omega x_0 - 3\dot{y}_0$$

$$\dot{y}(t_f) = (-660 \cdot 7.2722 \times 10^{-5} + 4 \times (-3.7242 \times 10^{-4}))\sqrt{3}/2$$

$$- 2 \cdot 0.02263/2 + 660 \times 7.2722 \times 10^{-5} + 3 \times 3.7242 \times 10^{-4}$$

$$\approx -0.01637 \text{ km/s} = -16.3725 \text{ m/s}$$

$$\dot{z}(t) = \dot{z}_0 \cos(\omega t) - \omega z_0 \sin(\omega t)$$

$$\dot{z}(t_f) = -1.2596 \times 10^{-3} \times \sqrt{3}/2 - 7.2722 \times 10^{-5} \cdot 10/2 \approx -1.4545 \times 10^{-3} \text{ km/s}$$

$$\Rightarrow \vec{D}_f = (\dot{x}(t_f) - \dot{x}_f)\vec{i} + (\dot{y}(t_f) - \dot{y}_f)\vec{j} + (\dot{z}(t_f) - \dot{z}_f)\vec{k} \Rightarrow \Delta \vec{D}_f = 0.01795 \text{ km/s}$$

$$= (7.2266 \times 10^{-3}\vec{i} - 0.01637\vec{j} - 1.4545 \times 10^{-3}\vec{k}) \text{ km/s}$$

#2.
$$\begin{cases} I_x \dot{\omega}_x = (I_y - I_z) \omega_y \omega_z + M_x \\ I_y \dot{\omega}_y = (I_z - I_x) \omega_x \omega_z + M_y \\ I_z \dot{\omega}_z = (I_x - I_y) \omega_x \omega_y + M_z \end{cases}$$

A. $\omega_x = 0, \omega_y = 0, \omega_z = \bar{\omega}_z = \text{const} \Rightarrow \dot{\omega}_z = 0, \forall t \geq 0$

$$\Rightarrow \begin{cases} I_x \dot{\omega}_x = 0 \\ I_y \dot{\omega}_y = 0 \end{cases} \Rightarrow \omega_x \text{ and } \omega_y \text{ are constant.}$$

$$\Rightarrow \omega_x = \omega_y = 0, \forall t \geq 0$$

$$\Rightarrow I_z \dot{\omega}_z = 0.$$

Since $\dot{\omega}_x = \dot{\omega}_y = \dot{\omega}_z = 0$, $\omega_x = 0 = \omega_y$, $\omega_z = \bar{\omega}_z$ is an equilibrium.

B. Let $\omega_x = 0 + \Delta\omega_x, \omega_y = 0 + \Delta\omega_y, \omega_z = \bar{\omega}_z + \Delta\omega_z$

$$\Rightarrow \begin{cases} I_x (\dot{0} + \Delta\dot{\omega}_x) = (I_y - I_z) (0 + \Delta\omega_y) (\bar{\omega}_z + \Delta\omega_z) + M_x \\ I_y (\dot{0} + \Delta\dot{\omega}_y) = (I_z - I_x) (0 + \Delta\omega_x) (\bar{\omega}_z + \Delta\omega_z) + M_y \\ I_z (\dot{\bar{\omega}_z} + \Delta\dot{\omega}_z) = (I_x - I_y) (0 + \Delta\omega_x) (0 + \Delta\omega_y) + M_z \end{cases}$$

$$\Rightarrow \begin{cases} I_x \Delta\dot{\omega}_x = (I_y - I_z) \Delta\omega_y \bar{\omega}_z + M_x \\ I_y \Delta\dot{\omega}_y = (I_z - I_x) \bar{\omega}_z \Delta\omega_x + M_y \\ I_z \Delta\dot{\omega}_z = 0 \end{cases}$$

C.
$$\begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix} \begin{bmatrix} \Delta\dot{\omega}_x \\ \Delta\dot{\omega}_y \\ \Delta\dot{\omega}_z \end{bmatrix} = \begin{bmatrix} 0 & (I_y - I_z) \bar{\omega}_z & 0 \\ (I_z - I_x) \bar{\omega}_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\omega_x \\ \Delta\omega_y \\ \Delta\omega_z \end{bmatrix} + M.$$

$$\Rightarrow \begin{bmatrix} \Delta\dot{\omega}_x \\ \Delta\dot{\omega}_y \\ \Delta\dot{\omega}_z \end{bmatrix} = \begin{bmatrix} 0 & (I_y - I_z) \bar{\omega}_z / I_x & 0 \\ (I_z - I_x) \bar{\omega}_z / I_y & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\omega_x \\ \Delta\omega_y \\ \Delta\omega_z \end{bmatrix} + M.$$

$$\Rightarrow \text{Characteristic equation: } s(s^2 - \frac{(I_y - I_z)(I_z - I_x)}{I_x I_y} \bar{\omega}_z^2) = 0.$$

$$\Rightarrow \text{Eigenvalues: } \lambda = 0 \text{ OR } \lambda = \pm \sqrt{\frac{(I_y - I_z)(I_z - I_x)}{I_x I_y}} \bar{\omega}_z.$$

If $I_y > I_z > I_x$ OR $I_y < I_z < I_x$, then all eigenvalues are real numbers.

There exist real positive eigenvalue, thus the system is unstable.

If $I_y = I_z$ OR $I_z = I_x$, then all eigenvalues are zero.

Hence the system is unstable.

If $I_y > I_z$ & $I_z < I_x$ OR $I_y < I_z$ & $I_z > I_x$, then there exist 2 complex eigenvalues, then the system is marginally stable.

(In this case, I_y can be same as I_x).

#3. $I_y \ddot{\theta} = 3\omega_0^2 (I_z - I_x) \Delta\theta + M_y.$

A. $100 \ddot{\theta} = 3\omega_0^2 (200 - 100) \Delta\theta \Rightarrow \ddot{\theta} = 1.5 \omega_0^2 \Delta\theta.$

$$\Rightarrow \begin{bmatrix} \dot{\Delta\theta} \\ \Delta\theta \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1.5 \omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta\theta \\ \dot{\Delta\theta} \end{bmatrix} \Rightarrow \text{eigenvalues } \lambda = \pm \sqrt{\frac{3}{2}} \omega_0.$$

Since there exist real positive eigenvalue, then the system is unstable whatever initial state is.

B. Let $M_y = -k_p \Delta\theta - k_d \dot{\Delta\theta}$

$\Rightarrow I_y \ddot{\Delta\theta} = 3\omega_0^2 (-I_x + I_z) \Delta\theta + M_y$

$\Rightarrow 100 \ddot{\Delta\theta} = 150 \omega_0^2 \Delta\theta - k_p \Delta\theta - k_d \dot{\Delta\theta}$

$\Rightarrow \ddot{\Delta\theta} + 0.01 k_d \dot{\Delta\theta} + (0.01 k_p - 1.5 \omega_0^2) \Delta\theta = 0$

$\Rightarrow \{s^2 + 0.01 k_d s + (0.01 k_p - 1.5 \omega_0^2)\} \Delta\theta_s = 0$

$\Rightarrow 0.01 k_d = 0.0112 \quad \& \quad 0.01 k_p - 1.5 \omega_0^2 = 0.000064$

$\therefore k_d = 1.12 \quad \& \quad k_p = 6.616 \times 10^{-3}$

$\therefore M_y = -6.616 \times 10^{-3} \Delta\theta - 1.12 \dot{\Delta\theta}$

#4 Let $\mathbf{z}^T = [z_1, z_2, z_3]$

$\Rightarrow \mathbf{z}^T \mathbf{z} + \dot{\mathbf{z}}^T \mathbf{z}$

$\frac{d}{dt}(\mathbf{z}^T \mathbf{z} + \dot{\mathbf{z}}^T \mathbf{z}) = 2\mathbf{z}^T \dot{\mathbf{z}} + 2\dot{\mathbf{z}}^T \mathbf{z} = \mathbf{z}^T S(\omega) \mathbf{z} + \mathbf{z}^T \dot{\mathbf{z}} + \dot{\mathbf{z}}^T \mathbf{z} - \mathbf{z}^T \dot{\mathbf{z}}$

$= \mathbf{z}^T S(\omega) \mathbf{z} = \mathbf{z} \cdot (\omega \times \mathbf{z}) = \det([\mathbf{z}, \omega, \mathbf{z}]) = 0$

$\therefore (a \times b) \cdot c = \det([a, b, c])$

Hence $\mathbf{z}^T \mathbf{z} + \dot{\mathbf{z}}^T \mathbf{z}$ is conserved.

Since $\mathbf{z}^T \mathbf{z} + \dot{\mathbf{z}}^T \mathbf{z} = 1$ at $t=0$, then $\mathbf{z}^T \mathbf{z} + \dot{\mathbf{z}}^T \mathbf{z} = 1 \quad \forall t \geq 0$

#1. $m = 100 \text{ kg}$. $r = 6578 \text{ km} \rightarrow \omega = \sqrt{228600/6578^3} \approx 1.1834 \times 10^{-3} \text{ /sec.}$

a. Let $\vec{r} = [x, y, z]^T$ and $\vec{F} = [F_x, F_y, F_z]^T$.

Then $m\ddot{\vec{r}} = \vec{F} \Rightarrow \ddot{\vec{r}} = \vec{F}/m$.

$$\Rightarrow \dot{\vec{r}} = (\vec{F}/m)t + \dot{\vec{r}}_0$$

$$\Rightarrow \vec{r} = \frac{1}{2}(\vec{F}/m)t^2 + \dot{\vec{r}}_0 t + \vec{r}_0$$

$$\Rightarrow \begin{cases} x(t) = \frac{1}{2}(F_x/m)t^2 + \dot{x}_0 t + x_0 \\ y(t) = \frac{1}{2}(F_y/m)t^2 + \dot{y}_0 t + y_0 \\ z(t) = \frac{1}{2}(F_z/m)t^2 + \dot{z}_0 t + z_0 \end{cases}$$

If $\vec{F} = 0$, then $\begin{cases} x(t) = \dot{x}_0 t + x_0 \\ y(t) = \dot{y}_0 t + y_0 \\ z(t) = \dot{z}_0 t + z_0 \end{cases} \dots (A)$

Now consider the analytic solutions of CW equations.

If $\omega t \ll 1$, $\sin \omega t \approx \omega t$ and $\cos \omega t \approx 1$.

$$\Rightarrow \begin{cases} x(t) = (\dot{x}_0/\omega)\omega t - (3x_0 + 2\dot{y}_0/\omega) + 4x_0 + 2\dot{y}_0 t/\omega \\ y(t) = (6x_0 + 4\dot{y}_0/\omega)\omega t + (2\dot{x}_0/\omega) - (6x_0 + 3\dot{y}_0)t + y_0 - 2\dot{x}_0 t/\omega \\ z(t) = (\dot{z}_0/\omega)\omega t + z_0 \end{cases}$$

$$\Rightarrow \begin{cases} x(t) = \dot{x}_0 t + x_0 \\ y(t) = \dot{y}_0 t + y_0 \\ z(t) = \dot{z}_0 t + z_0 \end{cases} \text{ which is same as (A)}$$

Thus, the approximations of sine and cosine should be within accuracy of about 3%.

$$\begin{cases} \omega t - \sin \omega t \leq 0.03 \sin \omega t \\ 1 - \cos \omega t \leq 0.03 \cos \omega t \end{cases} \Rightarrow \begin{cases} \omega t \leq 1.03 \sin \omega t \dots (B) \\ 1 \leq 1.03 \cos \omega t \dots (C) \end{cases}$$

From (C), $(1.03)^{-1} \leq \cos \omega t \Rightarrow \omega t \leq 0.2419$.

Since $1.03 \sin(0.2419) = 0.2468 \geq 0.2419$, then $\omega t \leq 0.2419$ is satisfying both (B) and (C).

Also, for satisfying approximated formula with in accuracy of about 3%, ωt should be less than or equal to 3% of period of trigonometry function $\Rightarrow \omega t \leq 0.03(2\pi) = 0.1885$.

Hence $\omega t \leq 0.1885 \Rightarrow t \leq 0.1885/\omega \approx 102.7784 \text{ sec.}$

Thus the maneuver time period is 102.7784 sec.

b. $\Delta \vec{r}(0) = -10\hat{z} \text{ km}$. $\Delta \dot{\vec{r}}(0) = 0$. $\Delta \vec{r}(60) = 0 \text{ km} \Rightarrow \Delta \dot{\vec{r}}(60) = 0$.

$$\begin{array}{cccc} t=0 & t=0^+ & \longrightarrow & t=60^- & t=60 \\ \left(\begin{array}{l} \Delta \vec{r}(0) = -10\hat{z} \\ \Delta \dot{\vec{r}}(0) = 0 \end{array} \right) & \left(\begin{array}{l} \Delta \vec{r}(0) = 0 \\ \Delta \dot{\vec{r}}(0) = \Delta \vec{V} \end{array} \right) & & \left(\begin{array}{l} \Delta \vec{r}(60^-) = \Delta \vec{r}(0) \cdot 60 + \Delta \vec{r}(0) \\ \Delta \dot{\vec{r}}(60^-) = \Delta \vec{V} \end{array} \right) & \left(\begin{array}{l} \Delta \vec{r}(60) = 0 \\ \Delta \dot{\vec{r}}(60) = 0 \end{array} \right) \end{array}$$

$$\Rightarrow \Delta \vec{r}(0) \cdot 60 + \Delta \vec{r}(0) = 0$$

$$\Rightarrow \Delta \vec{V} \cdot 60 - 10\hat{z} = 0$$

$$\Rightarrow \Delta \vec{V} = \frac{1}{6}\hat{z}$$

Thus, the initial $\Delta \vec{V}_i = \Delta \vec{V} = (1/6)\hat{z}$.

the final $\Delta \vec{V}_f = -\Delta \vec{V} = (-1/6)\hat{z}$.

#2. $L_2: (\bar{x}, \bar{y}, \bar{z}) = (-444645.6, 0, 0) \text{ km}$.

a. Let $x = \bar{x} + \Delta x$, $y = \Delta y$, $z = \Delta z$.

$$\begin{aligned} r_E^3 &= ((x - D_E)^2 + y^2 + z^2)^{-3/2} \\ &= ((\bar{x} + \Delta x - D_E)^2 + \Delta y^2 + \Delta z^2)^{-3/2} \\ &= ((\bar{x} - D_E)^2 + 2\Delta x(\bar{x} - D_E) + \Delta x^2)^{-3/2} \\ &= ((\bar{x} - D_E)^2 + 2\Delta x(\bar{x} - D_E))^{-3/2} \\ &= |\bar{x} - D_E|^{-3} \left(1 + 2\Delta x / (\bar{x} - D_E)\right)^{-3/2} \quad \left(\because \left|\frac{\Delta x}{\bar{x} - D_E}\right| \ll 1\right) \\ &= |\bar{x} - D_E|^{-3} \left(1 - 3\Delta x / (\bar{x} - D_E)\right) \end{aligned}$$

$$\begin{aligned} r_M^3 &= ((x + D_M)^2 + y^2 + z^2)^{-3/2} \\ &= ((\bar{x} + \Delta x + D_M)^2 + \Delta y^2 + \Delta z^2)^{-3/2} \\ &= ((\bar{x} + D_M)^2 + 2\Delta x(\bar{x} + D_M) + \Delta x^2)^{-3/2} \\ &= |\bar{x} + D_M|^{-3} \left(1 + 2\Delta x / (\bar{x} + D_M)\right)^{-3/2} \quad \left(\because \left|\frac{\Delta x}{\bar{x} + D_M}\right| \ll 1\right) \\ &= |\bar{x} + D_M|^{-3} \left(1 - 3\Delta x / (\bar{x} + D_M)\right) \end{aligned}$$

① $\ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\frac{\mu_E(x - D_E)}{r_E^3} - \frac{\mu_M(x + D_E)}{r_M^3} + \frac{F_x}{m}$

Left side : $\ddot{x} + \Delta \ddot{x} - 2\omega_0(\dot{y} + \Delta \dot{y}) - \omega_0^2(\bar{x} + \Delta x)$
 $= \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 \bar{x} - \omega_0^2 \Delta x - 2\omega_0 \Delta \dot{y}.$

Right side : $-\frac{\mu_E(\bar{x} + \Delta x - D_E)}{r_E^3} - \frac{\mu_M(\bar{x} + \Delta x + D_E)}{r_M^3} + \frac{F_x}{m}$

$$\begin{aligned} &= -\frac{\mu_E(\bar{x} - D_E + \Delta x)}{|\bar{x} - D_E|^3} \left(1 - 3\frac{\Delta x}{\bar{x} - D_E}\right) - \frac{\mu_M(\bar{x} + D_E + \Delta x)}{|\bar{x} + D_M|^3} \left(1 - 3\frac{\Delta x}{\bar{x} + D_M}\right) + \frac{F_x}{m} \\ &= -\frac{\mu_E(\bar{x} - D_E)}{|\bar{x} - D_E|^3} - \frac{4\mu_E \Delta x}{|\bar{x} - D_E|^3} - \frac{\mu_M(\bar{x} + D_M)}{|\bar{x} + D_M|^3} - \frac{4\mu_M \Delta x}{|\bar{x} + D_M|^3} + \frac{F_x}{m} \\ &= -\omega_0^2 \bar{x} - 4\omega_0^2 \Delta x + F_x/m. \end{aligned}$$

$\Rightarrow \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 \bar{x} - \omega_0^2 \Delta x - 2\omega_0 \Delta \dot{y} = -\omega_0^2 \bar{x} - 4\omega_0^2 \Delta x + F_x/m.$

$\Rightarrow \ddot{x} - 2\omega_0 \dot{y} + 3\omega_0^2 \Delta x = F_x/m.$

② $\ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m}$

Left side : $\ddot{y} + 2\omega_0(\dot{x} + \Delta \dot{x}) - \omega_0^2(\Delta y) = \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 \Delta y.$

Right side : $-\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m}$

$$\begin{aligned} &= -\frac{\mu_E \Delta y}{|\bar{x} - D_E|^3} \left(1 - 3\frac{\Delta x}{\bar{x} - D_E}\right) - \frac{\mu_M \Delta y}{|\bar{x} + D_M|^3} \left(1 - 3\frac{\Delta x}{\bar{x} + D_M}\right) + \frac{F_y}{m} \\ &= -\frac{\mu_E \Delta y}{|\bar{x} - D_E|^3} - \frac{\mu_M \Delta y}{|\bar{x} + D_M|^3} + \frac{F_y}{m} = -\omega_0^2 \Delta y + \frac{F_y}{m}. \end{aligned}$$

$\Rightarrow \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 \Delta y = -\omega_0^2 \Delta y + F_y/m.$

$\Rightarrow \ddot{y} + 2\omega_0 \dot{x} = F_y/m.$

③ $\ddot{z} = -\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m}$

Left side : $\ddot{z}$

Right side $-\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m}$

$$= -\frac{\mu_E \Delta z}{|\vec{r}-\vec{r}_E|^3} \left\{ 1 - 3 \frac{\vec{r} \cdot \vec{r}_E}{|\vec{r}-\vec{r}_E|^2} \right\} - \frac{\mu_M \Delta z}{|\vec{r}+\vec{r}_M|^3} \left\{ 1 - 3 \frac{\vec{r} \cdot \vec{r}_M}{|\vec{r}+\vec{r}_M|^2} \right\} + \frac{F_z}{m}$$

$$= -\frac{\mu_E \Delta z}{|\vec{r}-\vec{r}_E|^3} - \frac{\mu_M \Delta z}{|\vec{r}+\vec{r}_M|^3} + \frac{F_z}{m} = -\omega_0^2 z + \frac{F_z}{m}$$

$$\Rightarrow \ddot{\Delta z} + \omega_0^2 \Delta z = F_z/m \dots (A)$$

b. From (A), $\begin{bmatrix} \ddot{\Delta z} \\ \dot{\Delta z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta z \\ \dot{\Delta z} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{F_z}{m}$

Characteristic eqn, $\lambda^2 + \omega_0^2 = 0$

$\therefore$ eigenvalues : $\lambda = \pm i\omega_0$

c. Since eigenvalues are pure imaginary $\Rightarrow$ then out of plane motion is marginally stable.

#3 $H = \int_B \vec{r}(m) \times \vec{v}(m) dm$

$$= \int_B \vec{r} \times (\vec{\omega} \times \vec{r}) dm$$

$$= \int_B (\vec{r}^T \vec{r} - \vec{r} \vec{r}^T) \vec{\omega} dm$$

$$= \int_B \begin{bmatrix} r_y^2 + r_z^2 & -r_x r_y & -r_x r_z \\ -r_x r_y & r_x^2 + r_z^2 & -r_y r_z \\ -r_x r_z & -r_y r_z & r_x^2 + r_y^2 \end{bmatrix} dm \vec{\omega} = \vec{I} \vec{\omega}$$

By Coriolis thm,

$$\left(\frac{dH}{dt} \right)_{\text{inertial}} = \left(\frac{dH}{dt} \right)_{\text{body}} + \vec{\omega} \times H$$

$$\Rightarrow \dot{H} = (\vec{I} \dot{\vec{\omega}})' + \vec{\omega} \times H$$

$$\Rightarrow \vec{I} \dot{\vec{\omega}} = \vec{\omega} \times \vec{I} \vec{\omega}$$

$$\therefore \vec{I} \dot{\vec{\omega}} = \vec{\omega} \times \vec{I} \vec{\omega} + \vec{M}$$

Since $\vec{I} = \text{diag}(A, B, C)$, then

$$\begin{cases} A \dot{\omega}_1 = (B-C) \omega_2 \omega_3 + M_1 \\ B \dot{\omega}_2 = (C-A) \omega_1 \omega_3 + M_2 \\ C \dot{\omega}_3 = (A-B) \omega_1 \omega_2 + M_3 \end{cases}$$

b. Let $T = \frac{1}{2} \vec{\omega}^T \vec{I} \vec{\omega}$ which is the kinetic energy.

$$\frac{dT}{dt} = \frac{d}{dt} \left(\frac{1}{2} \vec{\omega}^T \vec{I} \vec{\omega} \right) = \vec{\omega}^T \dot{\vec{I}} \vec{\omega} = \vec{\omega}^T (\vec{\omega} \times \vec{I} \vec{\omega}) = \vec{\omega} \cdot (\vec{\omega} \times \vec{I} \vec{\omega})$$

$$= \det \begin{pmatrix} \omega_1 & \omega_2 & \omega_3 \\ \omega_1 & \omega_2 & \omega_3 \\ A\omega_1 & B\omega_2 & C\omega_3 \end{pmatrix} = 0 \therefore T \text{ is conserved.}$$

Let $H = |\vec{H}| \neq 0$.

$$\frac{d}{dt} H^2 = 2\vec{H} \cdot \dot{\vec{H}} = 2(\vec{I}\vec{\omega})^T (\dot{\vec{I}}\vec{\omega}) = 2(\vec{I}\vec{\omega})^T (\vec{\omega} \times \vec{I}\vec{\omega})$$

$$= \det \begin{bmatrix} A\omega_1 & B\omega_2 & C\omega_3 \\ \dot{A}\omega_1 & \dot{B}\omega_2 & \dot{C}\omega_3 \end{bmatrix} = 0.$$

Thus the magnitude of angular momentum H is conserved.

#Q. Same as other Final exam.

I never received nor gave help regarding this exam

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$$m = 100 \text{ kg} \quad r = 6578 \text{ km}$$

$$m \Delta \ddot{x} = F_x$$

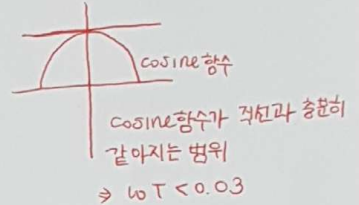
$$m \Delta \ddot{y} = F_y$$

$$m \Delta \ddot{z} = F_z$$

$$\sin \omega t \approx \omega t$$

$$\cos \omega t \approx 1$$

$$\Rightarrow 0.03 \text{ 이내}$$



a. 충분히 작은 시간 t 에 대해서 $\sin(\omega t) \approx \omega t$, $\cos \omega t \approx 1$ 로 근사된다.

즉 주어진 C-W 방정식은 다음과 같이 근사된다.

$$x(t) = \frac{\dot{x}_0}{\omega} \omega t - \left(3x_0 + \frac{2\dot{y}_0}{\omega}\right) \cdot 1 + \left(4x_0 + \frac{2\dot{y}_0}{\omega}\right) = x_0 + \dot{x}_0 t$$

$$y(t) = \left(6x_0 + \frac{4\dot{y}_0}{\omega}\right) \omega t + \frac{2\dot{x}_0}{\omega} \cdot 1 - \left(6x_0 + 3\dot{y}_0\right) t + \left(y_0 - \frac{2\dot{x}_0}{\omega}\right)$$

$$= y_0 + \dot{y}_0 t$$

$$z(t) = \frac{\dot{z}_0}{\omega} \omega t + z_0 \cdot 1 = z_0 + \dot{z}_0 t$$

... ①

$$m \Delta \ddot{x} = F_x$$

$m \Delta \ddot{y} = F_y$ 가 위의 식 ①의 3% 이내 정확도를 가진 approximation 이어야 한다.

$$m \Delta \ddot{z} = F_z$$

$$\text{식 ①에서 perturbation만 고려하면} \quad \begin{aligned} \Delta x &= \dot{x}_0 t \\ \Delta y &= \dot{y}_0 t \\ \Delta z &= \dot{z}_0 t \end{aligned} \quad \dots ②$$

$$\begin{aligned} \Delta \ddot{x} &= \frac{F_x}{m} & \Delta x &= \frac{F_x}{m} t^2 + \dot{x}_0 t \\ \Delta \ddot{y} &= \frac{F_y}{m} & \Delta y &= \frac{F_y}{m} t^2 + \dot{y}_0 t \\ \Delta \ddot{z} &= \frac{F_z}{m} & \Delta z &= \frac{F_z}{m} t^2 + \dot{z}_0 t \end{aligned} \quad \dots ③$$

③과 ②의 차이가 ②의 0.03 이내여야 하므로

$$\frac{F_x}{m} t^2 < \dot{x}_0 t \times 0.03 \Rightarrow T < \frac{0.03 \dot{x}_0 m}{F_x}$$

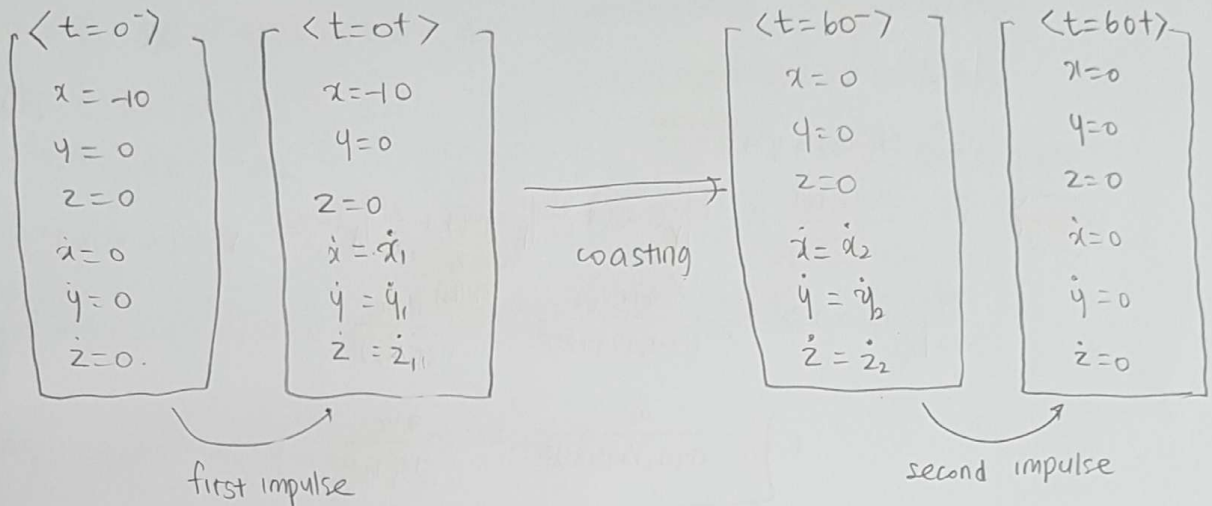
마찬가지로 y 와 z 에 대해서도

$$T < \frac{0.03 \dot{y}_0 m}{F_y} \quad T < \frac{0.03 \dot{z}_0 m}{F_z}$$

x, y, z 에 대한 세개의 부등식을 모두 만족시키는 T 를 찾아야 한다.

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b.



coasting 부분에서 CW 방향성을 따라 전파되므로

$$x(60) = x(0) + \dot{x}_0 \times 60 \text{ sec}$$

$$\Rightarrow 0 \text{ km} = -10 \text{ km} + (\dot{x}_1) \times 60 \text{ sec}$$

$$\Rightarrow \dot{x}_1 = \frac{1}{6} \text{ km/sec}$$

y축과 z축방향으로는 위치, 속도 상의 변화가 없으며 식①과 같이 근사했을 때 $\dot{x}, \dot{y}, \dot{z}$ 는 coupled되지 않았으므로 y축이나 z축 방향으로는 impulse를 가해줄 필요가 없다.

또한 문제에서

$$x(t) = x_0 + \dot{x}_0 t \Rightarrow \dot{x}(t) = \dot{x}_0 \text{ 이다.}$$

60초 후에 second impulse 적용의 x방향 속도는 다음과 같다.

$$\dot{x}(60) = \dot{x}_2 = \dot{x}_0 = \dot{x}_1 = \frac{1}{6} \text{ km/sec}$$

따라서 y축이나 z축방향으로는 impulse 가할 필요가 없으므로 첫번째, 두번째 순간 속력은 다음과 같다

$$\Delta \dot{x}_1 = \dot{x}_1 - 0 = \left(\frac{1}{6} \text{ km/sec}\right) i = (0.1667 \text{ km/s}) i$$

$$\Delta \dot{x}_2 = 0 - \dot{x}_2 = \left(-\frac{1}{6} \text{ km/sec}\right) i = (-0.1667 \text{ km/s}) i$$

$$\begin{aligned} \text{총 요구되는 impulse는 } \Delta \dot{x}_{\text{tot}} &= |\Delta \dot{x}_1| + |\Delta \dot{x}_2| = 0.1667 \text{ km/s} + 0.1667 \text{ km/s} \\ &= 0.3334 \text{ km/s} \end{aligned}$$

$$r_e^2 = (x - D_E)^2 + y^2 + z^2$$

단항화

$$f = -\frac{M_E(x - D_E)}{r_e} \approx \frac{\partial f}{\partial x} \Big|_{x_e} (x - x_e) + \frac{\partial f}{\partial y} \Big|_{x_e} (y - y_e) + \frac{\partial f}{\partial z} \Big|_{x_e} (z - z_e) \quad x_e = [x_e \ y_e \ z_e]^T$$

$$\frac{\partial f}{\partial x} \Big|_{x_e} (x - x_e) \Rightarrow + \frac{3}{2} \frac{M_E(x - D_E)^2}{((x - D_E)^2 + y^2 + z^2)^{\frac{3}{2}}} \xrightarrow{\text{대입 (L2)}} \frac{3M_E}{(x_e - D_E)^3}$$

$$+ \left[-\frac{M_E}{((x - D_E)^2 + y^2 + z^2)^{\frac{3}{2}}} = -\frac{M_E}{(x_e - D_E)^3} \right]$$

$$\frac{2M_E}{(x_e - D_E)^3}$$

$$\frac{\partial f}{\partial y} \Big|_{x_e} (y - y_e) \Rightarrow + \frac{3}{2} \frac{M_E(x - D_E)2y}{((x - D_E)^2 + y^2 + z^2)^{\frac{3}{2}}} \xrightarrow{\text{대입 (L2)}} 0$$

$$\therefore f \approx \frac{\partial f}{\partial x} \Big|_{x_e} (x - x_e) + \cancel{\frac{\partial f}{\partial y} \Big|_{x_e} (y - y_e)} + \cancel{\frac{\partial f}{\partial z} \Big|_{x_e} (z - z_e)} \quad x_e = [x_e \ y_e \ z_e]^T$$

$$= \frac{2M_E}{(x_e - D_E)^3} (x - x_e)$$

절대값의 삼승으로 계산해야 함.
음수화면 틀려. 계산주의

2 a. $x = \bar{x} + \Delta x = -444645.6 + \Delta x$

$$y = \bar{y} + \Delta y = \Delta y$$

$$z = \bar{z} + \Delta z = \Delta z$$

방정식에서 내인항 $-\frac{M_E(x-D_E)}{r_E^3}$ 애나 등재2!

r_E^3 안 선형화하려면 안됨. 분자까지 해야해

$$-\frac{M_E(x-D_E)}{r_E^3} = g \Rightarrow \frac{\partial g}{\partial x} \Big|_{\text{등재2}} \text{ 편미분해야함.}$$

식을 주어진 equation of motion에 대입하면

← 이진페이지풀이참고.

$$\Delta \ddot{x} - 2\omega_0 \Delta \dot{y} - \omega_0^2 \Delta x = -\frac{M_E(-444645.6 + \Delta x - D_E)}{r_E^3} - \frac{M_M(-444645.6 + \Delta x + D_M)}{r_M^3}$$

$$\Delta \ddot{y} + 2\omega_0 \Delta \dot{x} - \omega_0^2 \Delta y = -\frac{M_E \Delta y}{r_E^3} - \frac{M_M \Delta y}{r_M^3} + \frac{F_y}{m}$$

$$\Delta \ddot{z} = -\frac{M_E \Delta z}{r_E^3} - \frac{M_M \Delta z}{r_M^3} + \frac{F_z}{m}$$

여기서 $r_E^{-3} = \left((-444645.6 + \Delta x - D_E)^2 + \Delta y^2 + \Delta z^2 \right)^{-\frac{3}{2}}$

$$= \left((-449319.6 + \Delta x)^2 \right)^{-\frac{3}{2}}$$

$$\approx -\frac{3}{2} \left(2.0189 \times 10^{11} - 2449319.6 \Delta x \right)$$

비슷하게 $r_M^{-3} = \left((-444645.6 + \Delta x + D_M)^2 + \Delta y^2 + \Delta z^2 \right)^{-\frac{3}{2}}$

$$= \left((-64572.6 + \Delta x)^2 \right)^{-\frac{3}{2}}$$

$$\approx -\frac{3}{2} \left(4.1696 \times 10^9 - 264572.6 \Delta x \right) \quad \text{3 선형화된다.}$$

이를 위식식에 대입하여 정리하면

$$\Delta \ddot{x} - 2\omega_0 \Delta \dot{y} + \left(-\omega_0^2 + M_E(-449319.6 + \Delta x) \left(-\frac{3}{2} \right) (2.0189 \times 10^{11} - 2449319.6 \Delta x) \right. \\ \left. + M_M(-64572.6 + \Delta x) \left(-\frac{3}{2} \right) (4.1696 \times 10^9 - 264572.6 \Delta x) \right) \Delta x = 0.$$

$$\Rightarrow \Delta \ddot{x} - (5.3240 \times 10^{-6} \text{ rad/sec}) \Delta \dot{y} + \left(-7.0862 \times 10^{-12} \text{ rad/sec}^2 - (597901.5 \text{ km}^3/\text{s}^2) (-9.0713 \times 10^{16} + 4.0378 \times 10^{11} \Delta x) \right. \\ \left. - (7330.5 \text{ km}^3/\text{s}^2) (-2.6924 \times 10^{14} + 8.3392 \times 10^9 \Delta x) \right) \Delta x = 0$$

$$\Rightarrow \Delta \ddot{x} - (5.3240 \times 10^{-6} \text{ rad/sec}) \Delta \dot{y} + (5.4239 \times 10^{22} \text{ cm/s}^2) \Delta x = 0$$

$$\Delta \ddot{y} + 2\omega_0 \Delta \dot{x} + \left(-\omega_0^2 + M_E \left(-\frac{3}{2} \right) (2.0189 \times 10^{11} - 2449319.6 \Delta x) + M_M \left(-\frac{3}{2} \right) (4.1696 \times 10^9 - 264572.6 \Delta x) \right) \Delta y = 0$$

$$\Rightarrow \Delta \ddot{y} + (5.3240 \times 10^{-6}) \Delta \dot{x} + \left(-7.0862 \times 10^{-12} - (597901.5) (2.0189 \times 10^{11}) - (7330.5) (4.1696 \times 10^9) \right) \Delta y = 0$$

$$\Rightarrow \Delta \ddot{y} + (5.3240 \times 10^{-6}) \Delta \dot{x} - 1.2074 \times 10^{17} \Delta y = 0$$

$$\Delta \ddot{z} + \left(M_E \left(-\frac{3}{2} \right) (-2.0189 \times 10^{11} + 449319.64x) + M_M \left(-\frac{3}{2} \right) (-4.1696 \times 10^9 + 64572.64x) \right) \Delta z = 0$$

$$\Rightarrow \Delta \ddot{z} + ((597901.5)(2.0189 \times 10^{11}) + (7330.5)(4.1696 \times 10^9)) \Delta z = 0$$

$$\Rightarrow \Delta \ddot{z} + 1.2072 \times 10^{18} \Delta z = 0$$

∴ in-plane linearized equation

$$\begin{bmatrix} \Delta \ddot{x} \\ \Delta \ddot{y} \end{bmatrix} + \begin{bmatrix} 0 & -5.3240 \times 10^{-5} \\ 5.3240 \times 10^{-5} & 0 \end{bmatrix} \begin{bmatrix} \Delta \dot{x} \\ \Delta \dot{y} \end{bmatrix} + \begin{bmatrix} 5.4239 \times 10^{22} & 0 \\ 0 & -1.2074 \times 10^{18} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = 0$$

out-of-plane linearized equation

$$\Delta \ddot{z} + 1.2072 \times 10^{18} \Delta z = 0$$

b. $\Delta \ddot{z} + 1.2072 \times 10^{18} \Delta z = 0$ 을 Laplace transform 취하면

$$s^2 + 1.2072 \times 10^{18} = 0$$

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$$s = \pm 1.0988 \times 10^9 i$$

따라서 eigenvalue 두개가 모두 순허수이다.

c. eigenvalue가 순허수만 존재하는 경우 안정성을 판단할 수 없다. 시스템의 안정성을 판단하기 위해서는 주어진 비선형 방정식에서 Lyapunov function을 찾는 등의 다른 방법을 활용해야 한다.

eigenvalue가 0인 경우 $\rightarrow \det(A - \lambda I) = 0$.

system 자체가 수렴되지 않는 경우다.

혹은 dynamic이 없는 경우

3 a. $H = I\omega$

$$\frac{d}{dt}(H)^I = \frac{d}{dt}(I\omega)^I = \frac{d}{dt}(I\dot{\omega})^b + \vec{\omega} \times I\vec{\omega} \quad (\text{by Coriolis Theorem})$$

$$\Rightarrow M = I\ddot{\omega} + \vec{\omega} \times (I\dot{\omega}) \quad (\because \text{body-fixed frame or } \dot{I} = 0) \text{ in body-fixed frame.}$$

$$= I\ddot{\omega} + S(\vec{\omega})I\dot{\omega}$$

$$\Rightarrow \begin{bmatrix} M_1 \\ M_2 \\ M_3 \end{bmatrix} = \begin{bmatrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & C \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \begin{bmatrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & C \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

$$= \begin{bmatrix} A\dot{\omega}_1 \\ B\dot{\omega}_2 \\ C\dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} 0 & -B\omega_3 & C\omega_2 \\ A\omega_3 & 0 & -C\omega_1 \\ -A\omega_2 & B\omega_1 & 0 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

$$= \begin{bmatrix} A\dot{\omega}_1 + (-B+C)\omega_2\omega_3 \\ B\dot{\omega}_2 + (A-C)\omega_1\omega_3 \\ C\dot{\omega}_3 + (-A+B)\omega_1\omega_2 \end{bmatrix}$$

$$\Rightarrow A\dot{\omega}_1 = (B-C)\omega_2\omega_3 + M_1$$

$$B\dot{\omega}_2 = (C-A)\omega_1\omega_3 + M_2$$

$$C\dot{\omega}_3 = (A-B)\omega_1\omega_2 + M_3$$

Euler's equation of motion in body fixed frame은 위와 같다.

b. $M_1 = M_2 = M_3 = 0$ 인 경우에

$$T = \frac{1}{2} \vec{\omega}^T \mathbf{I} \vec{\omega} = \frac{1}{2} (A\omega_1^2 + B\omega_2^2 + C\omega_3^2)$$

$\frac{d}{dt}T = A\omega_1\dot{\omega}_1 + B\omega_2\dot{\omega}_2 + C\omega_3\dot{\omega}_3$ 여기서 $\dot{\omega}_1, \dot{\omega}_2, \dot{\omega}_3$ 에 Euler's equation 을 대입하면,

$$= \omega_1(B-C)\omega_2\omega_3 + \omega_2(C-A)\omega_1\omega_3 + \omega_3(A-B)\omega_1\omega_2$$

$$= \omega_1\omega_2\omega_3 (B/C - C/A + A/B)$$

$$= 0$$

$\therefore$ kinetic energy T 은 보존된다.

H 의 크기를 고려하기 위해 그것의 제곱을 생각하면,

$$H^2 = (\mathbf{I}\vec{\omega})^T (\mathbf{I}\vec{\omega}) = A^2\omega_1^2 + B^2\omega_2^2 + C^2\omega_3^2$$

$$\frac{d}{dt}(H^2) = 2A^2\omega_1\dot{\omega}_1 + 2B^2\omega_2\dot{\omega}_2 + 2C^2\omega_3\dot{\omega}_3 \quad (\text{body fixed frame 에서 } \dot{A} = \dot{B} = \dot{C} = 0)$$

$$= 2A\omega_1(B-C)\omega_2\omega_3 + 2B\omega_2(C-A)\omega_1\omega_3 + 2C\omega_3(A-B)\omega_1\omega_2$$

$$= 2\omega_1\omega_2\omega_3 (A/B - A/C + B/C - A/B + C/A - B/C) = 0$$

$\Rightarrow H\dot{H} = 0$ H 가 0이 아닌 경우에 $\dot{H} = 0$, $H=0$ 인 경우는 equilibrium point 이므로 상관없다.

$\therefore$ Angular momentum 의 크기는 보존된다.

$$\boxed{4} \quad I_y \ddot{\Delta\theta} = 3\omega_0^2 (-I_x + I_z) \Delta\theta + M_y$$

$$\omega_0 = 0.0012 \text{ rad/sec}$$

$$I_x = 150 \text{ kg} \cdot \text{m}^2, \quad I_y = 100 \text{ kg} \cdot \text{m}^2, \quad I_z = 200 \text{ kg} \cdot \text{m}^2$$

$$a. \quad \ddot{\Delta\theta} - 3\omega_0^2 \frac{-I_x + I_z}{I_y} \Delta\theta = 0 \quad (\Delta\theta, \dot{\Delta\theta}) = (0, 0) : \text{equilibrium}$$

$$\ddot{\Delta\theta} - 3(0.0012 \text{ rad/s})^2 \frac{-(150+200)}{100} \Delta\theta = 0$$

$$\ddot{\Delta\theta} - 2.16 \times 10^{-6} \Delta\theta = 0$$

$$\Rightarrow S^2 - 2.16 \times 10^{-6} = 0$$

$$\Rightarrow S = \pm 1.4697 \times 10^{-3}$$

eigenvalue 중에 양수인 값이 존재하므로 이 equilibrium point는 unstable 하다.

b. linear state feedback control (PD control)

$$\begin{aligned} M_y &= -K_p(\Delta\theta - \Delta\theta_c) - K_d(\dot{\Delta\theta} - \dot{\Delta\theta}_c) \\ &= -K_p \Delta\theta - K_d \dot{\Delta\theta} \quad (\because \Delta\theta_c = \dot{\Delta\theta}_c = 0) \end{aligned}$$

$$\ddot{\Delta\theta} = 3\omega_0^2 \frac{-I_x + I_z}{I_y} \Delta\theta + \frac{M_y}{I_y} \quad \text{여기서 } M_y \text{를 대입}$$

$$= 3\omega_0^2 \frac{-I_x + I_z}{I_y} \Delta\theta + \frac{-K_p}{I_y} \Delta\theta + \frac{-K_d}{I_y} \dot{\Delta\theta}$$

$$\Rightarrow \ddot{\Delta\theta} + \frac{K_d}{I_y} \dot{\Delta\theta} + \left(-3\omega_0^2 \frac{I_z - I_x}{I_y} + \frac{K_p}{I_y} \right) \Delta\theta = 0$$

양변을 Laplace Transform 주하면

$$S^2 + \frac{K_d}{I_y} S + \left(-3\omega_0^2 \frac{I_z - I_x}{I_y} + \frac{K_p}{I_y} \right) = 0 \quad \text{이것이 문제에서 주어진 } S^2 + 0.0512S + 0.000064 = 0 \text{ 와}$$

같아야 하므로

$$\frac{K_d}{I_y} = 0.0112$$

$$\Rightarrow K_d = I_y \cdot 0.0112 = 100 \times 0.0112 = 1.12$$

$$-3\omega_0^2 \frac{I_z - I_x}{I_y} + \frac{K_p}{I_y} = 0.000064$$

$$\Rightarrow \frac{K_p}{I_y} = 0.000064 + 3(0.0012)^2 \frac{200-150}{100} = 0.000064 + 2.16 \times 10^{-6} = 6.616 \times 10^{-5}$$

$$\Rightarrow K_p = I_y \cdot (6.616 \times 10^{-5}) = 100 \times (6.616 \times 10^{-5}) = 6.616 \times 10^{-3}$$

$\therefore$ feedback gain은

$K_p = 6.616 \times 10^{-3}$, $K_d = 1.12$ 로 가해주면 주어진 equilibrium에서도 asymptotically stable해진다.

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1. $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} \Rightarrow \vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$
 By Newton's Law,

$$\frac{d^2\vec{r}}{dt^2} = -\frac{\mu}{r^3}\vec{r} + \vec{F} \Rightarrow \begin{cases} \ddot{x} = -\frac{\mu}{r^3}x + F_x \\ \ddot{y} = -\frac{\mu}{r^3}y + F_y \\ \ddot{z} = -\frac{\mu}{r^3}z + F_z \end{cases}$$

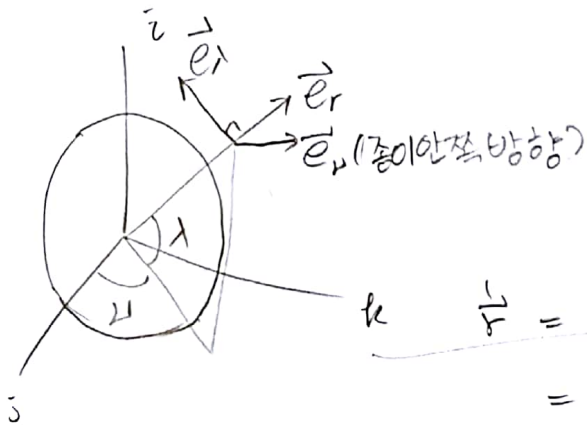
where $r = \sqrt{x^2 + y^2 + z^2}$

$$\Rightarrow \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{\mu}{r^3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{\mu}{r^3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\frac{\mu}{r^3} & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}$$

m? -2

3. Let $\vec{r} = r\vec{e}_r \Rightarrow \vec{F} = F_r\vec{e}_r + F_\nu\vec{e}_\nu + F_\lambda\vec{e}_\lambda$

$$\Rightarrow \vec{\omega} = \dot{\mu}\cos\lambda\vec{e}_\lambda + \dot{\nu}\sin\lambda\vec{e}_r - \dot{\lambda}\vec{e}_\nu$$



$$\begin{cases} \vec{e}_r = \vec{e}_\nu \times \vec{e}_\lambda \\ \vec{e}_\nu = \vec{e}_\lambda \times \vec{e}_r \\ \vec{e}_\lambda = \vec{e}_r \times \vec{e}_\nu \end{cases}$$

by Coriolis thm

$$\begin{aligned} \frac{1}{r} \vec{F} &= \ddot{r}\vec{e}_r + \dot{r}\dot{\vec{e}}_r = \ddot{r}\vec{e}_r + r\vec{\omega} \times \vec{e}_r \\ &= \ddot{r}\vec{e}_r + r\dot{\nu}\cos\lambda\vec{e}_\lambda \times \vec{e}_r - r\dot{\lambda}\vec{e}_\nu \times \vec{e}_r \\ &= \ddot{r}\vec{e}_r + r\dot{\nu}\cos\lambda\vec{e}_\nu + r\dot{\lambda}\vec{e}_\lambda \end{aligned}$$

$$\begin{aligned} \ddot{\vec{r}} &= \ddot{r}\vec{e}_r + (\dot{r}\dot{\nu}\cos\lambda + r\ddot{\nu}\cos\lambda - r\dot{\lambda}\dot{\nu}\sin\lambda)\vec{e}_\nu + (\dot{r}\dot{\lambda} + r\ddot{\lambda})\vec{e}_\lambda \\ &+ \dot{r}(\vec{\omega} \times \vec{e}_r) + r\dot{\nu}\cos\lambda(\vec{\omega} \times \vec{e}_\nu) + r\dot{\lambda}(\vec{\omega} \times \vec{e}_\lambda) \\ &= \ddot{r}\vec{e}_r + (\dot{r}\dot{\nu}\cos\lambda + r\ddot{\nu}\cos\lambda - r\dot{\lambda}\dot{\nu}\sin\lambda)\vec{e}_\nu + (\dot{r}\dot{\lambda} + r\ddot{\lambda})\vec{e}_\lambda \\ &+ (\dot{r}\dot{\nu}\cos\lambda\vec{e}_\lambda - \dot{r}\dot{\lambda}\vec{e}_\nu) \times \vec{e}_r \\ &+ (\dot{r}\dot{\nu}\cos^2\lambda\vec{e}_\lambda + r\dot{\nu}\cos\lambda\sin\lambda\vec{e}_r) \times \vec{e}_\nu \\ &+ (r\dot{\lambda}\dot{\nu}\sin\lambda\vec{e}_r - r\dot{\lambda}^2\vec{e}_\nu) \times \vec{e}_\lambda \end{aligned}$$

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$$\begin{aligned}
 &= \vec{e}_r (\ddot{r} - r\dot{\lambda}^2 \cos^2 \lambda - r\dot{\lambda}^2) + \vec{e}_\lambda (\dot{r}\dot{\lambda} + r\ddot{\lambda} + \dot{r}\dot{\lambda} + r\dot{\lambda}^2 \cos \lambda \sin \lambda) \\
 &\quad + (\dot{r}\dot{\lambda} \cos \lambda + r\ddot{\lambda} \cos \lambda - r\dot{\lambda}\dot{\lambda} \sin \lambda + \dot{r}\dot{\lambda} \cos \lambda - r\dot{\lambda}\dot{\lambda} \sin \lambda) \vec{e}_\nu \\
 &= (\ddot{r} - r\dot{\lambda}^2 \cos^2 \lambda - r\dot{\lambda}^2) \vec{e}_r + (r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\lambda}^2 \cos \lambda \sin \lambda) \vec{e}_\lambda \\
 &\quad + (r\ddot{\lambda} \cos \lambda + 2\dot{r}\dot{\lambda} \cos \lambda - 2r\dot{\lambda}\dot{\lambda} \sin \lambda) \vec{e}_\nu
 \end{aligned}$$

(2)

since $\frac{\mu}{r} = -\frac{\mu}{r^3} \vec{r} + \vec{F} = -\frac{\mu}{r^2} \vec{e}_r + \vec{F}$, then

$$\begin{cases}
 \ddot{r} - r\dot{\lambda}^2 \cos^2 \lambda - r\dot{\lambda}^2 = -\frac{\mu}{r^2} + F_r \\
 r\ddot{\lambda} \cos \lambda + 2\dot{r}\dot{\lambda} \cos \lambda - 2r\dot{\lambda}\dot{\lambda} \sin \lambda = F_\nu \\
 r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\lambda}^2 \cos \lambda \sin \lambda = F_\lambda
 \end{cases}$$

C. let $F = R$, $\vec{v} = \sqrt{\frac{\mu}{R^3}} t = \omega t$, $\vec{\lambda} = 0$. $\sin \theta \approx \theta$
 $\cos \theta \approx 1$

define $r = R + \Delta r$, $v = \omega t + \Delta v$, $\lambda = \Delta \lambda$.

from $\ddot{r} - r\dot{\lambda}^2 \cos^2 \lambda - r\dot{\lambda}^2 + \frac{\mu}{r^2} = F_r$,

$$\ddot{r} - (R + \Delta r)(\omega + \Delta \omega)^2 \cos^2 \Delta \lambda - (R + \Delta r) \dot{\lambda}^2 + \frac{\mu}{(R + \Delta r)^2} = F_r$$

$$\begin{aligned}
 &= \ddot{\Delta r} - (R + \Delta r)(\omega^2 + 2\omega\Delta\omega + \Delta\omega^2) \cos^2 \Delta \lambda - (R + \Delta r) \dot{\lambda}^2 + \frac{\mu}{(R + \Delta r)^2} \\
 &= \ddot{\Delta r} - R\omega^2 - 2\omega R\Delta\omega - \omega^2 \Delta r - 2\omega\Delta\omega \Delta r + \frac{\mu}{R^2} (1 - 2\frac{\Delta r}{R}) \\
 &= \ddot{\Delta r} - \cancel{\omega^2 R} - 2\omega R\Delta\omega - \omega^2 \Delta r + \cancel{\omega^2 R} - 2\omega^2 \Delta r \\
 &= \ddot{\Delta r} - 2\omega R\Delta\omega - 3\omega^2 \Delta r
 \end{aligned}$$

$\therefore \ddot{\Delta r} - 2\omega R\Delta\omega - 3\omega^2 \Delta r = F_r$

From $r\ddot{\lambda} \cos \lambda + 2\dot{r}\dot{\lambda} \cos \lambda - 2r\dot{\lambda}\dot{\lambda} \sin \lambda = F_\nu$

$$\begin{aligned}
 &\ddot{r} \cos \lambda + 2\dot{r}\dot{\lambda} \cos \lambda - 2r\dot{\lambda}\dot{\lambda} \sin \lambda = F_\nu \\
 &= (R + \Delta r) \ddot{\Delta \lambda} + 2\dot{\Delta r}(\omega + \Delta \omega) - 2(R + \Delta r)(\omega + \Delta \omega) \Delta \lambda \sin \Delta \lambda \\
 &= R\ddot{\Delta \lambda} + \Delta r \ddot{\Delta \lambda} + 2\omega \Delta r + 2\dot{\Delta r} \omega \\
 &= R\ddot{\Delta \lambda} + 2\omega \Delta r
 \end{aligned}$$

$R\ddot{\Delta \lambda} + 2\omega \Delta r = F_\nu$

Taylor expansion
OR
Binomial expansion
 $(1+x)^{-a} \approx 1 - ax$

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$$r''_{\lambda} + 2\dot{r}_{\lambda} + r\dot{\omega}^2 \cos\lambda \sin\lambda = F_{\lambda}$$

$$\begin{aligned} &\rightarrow (R+\Delta r) \Delta''_{\lambda} + 2(R+\Delta r) \dot{\Delta}_{\lambda} + (R+\Delta r)(\omega + \dot{\Delta}\omega)^2 \frac{\cos\Delta\lambda \sin\Delta\lambda}{\approx 1 \approx \Delta\lambda} \\ &= R\Delta''_{\lambda} + \Delta r \Delta''_{\lambda} + 2R\dot{\Delta}_{\lambda} + (R+\Delta r)(\omega^2 + 2\dot{\omega}\omega + \dot{\Delta}\omega^2) \Delta\lambda \\ &= R\Delta''_{\lambda} + (R+\Delta r)(\omega^2 + 2\dot{\omega}\omega) \Delta\lambda \\ &= R\Delta''_{\lambda} + R\omega^2 \Delta\lambda + \Delta r \omega^2 \Delta\lambda + 2R\dot{\omega}\omega \Delta\lambda + \Delta r 2\dot{\omega}\omega \Delta\lambda \\ &= R\Delta''_{\lambda} + \omega^2 R \Delta\lambda \end{aligned}$$

$$\therefore R\Delta''_{\lambda} + \omega^2 R \Delta\lambda = F_{\lambda}$$

let $x = \Delta r$, $y = R\dot{\omega}$, $z = R\Delta\lambda$, then

$$\begin{cases} \ddot{x} - 2\omega\dot{y} - 3\omega^2 x = F_r \\ \ddot{y} + 2\omega\dot{x} = F_y \\ \ddot{z} + \omega^2 z = F_{\lambda} \end{cases}$$

which are CW equations.

#2.

A. let $\omega_x = 0 = \omega_y$ & $\omega_z = \bar{\omega}_z = \text{constant}$, $M_x = 0 = M_y = M_z$.

$$\begin{aligned} I_x \dot{\omega}_x &= (I_y - I_z) \omega_y \omega_z + M_x \\ &= (I_y - I_z) 0 \cdot \bar{\omega}_z + 0 = 0 \end{aligned}$$

$$\begin{aligned} I_y \dot{\omega}_y &= (I_z - I_x) \omega_x \omega_z + M_y \\ &= (I_z - I_x) 0 \cdot \bar{\omega}_z + 0 = 0 \end{aligned}$$

$$\begin{aligned} I_z \dot{\omega}_z &= (I_x - I_y) \omega_x \omega_y + M_z \\ &= (I_x - I_y) 0 \cdot 0 + 0 = 0 \end{aligned}$$

$$\Rightarrow \begin{cases} I_x \dot{\omega}_x = 0 \\ I_y \dot{\omega}_y = 0 \\ I_z \dot{\omega}_z = 0 \end{cases}$$

I_x, I_y, I_z 가 0인 경우는 물리적으로 의미가 없는 문제에서는 나올 수 없으므로 고려하지 않음

Since $I_x > 0, I_y > 0, I_z > 0$ by definition, then

$$\omega_x = 0 = \omega_y = \omega_z$$

Thus it is an equilibrium solution.

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3. Let $\omega_x = \Delta\omega_x$, $\omega_y = \Delta\omega_y$, $\omega_z = \bar{\omega}_z + \Delta\omega_z$.

From $I_x \dot{\omega}_x = (I_y - I_z) \omega_y \omega_z + M_x$,

$$\begin{aligned} \Rightarrow M_x &= I_x \dot{\omega}_x - (I_y - I_z) \omega_y \omega_z \\ &= I_x \Delta \dot{\omega}_x - (I_y - I_z) \Delta \omega_y (\bar{\omega}_z + \Delta \omega_z) \\ &= I_x \Delta \dot{\omega}_x - (I_y - I_z) \Delta \omega_y \bar{\omega}_z - (I_y - I_z) \Delta \omega_y \Delta \omega_z \\ &= I_x \Delta \dot{\omega}_x - (I_y - I_z) \bar{\omega}_z \Delta \omega_y. \end{aligned}$$

From $I_y \dot{\omega}_y = (I_z - I_x) \omega_x \omega_z + M_y$,

$$\begin{aligned} \Rightarrow M_y &= I_y \dot{\omega}_y - (I_z - I_x) \omega_x \omega_z \\ &= I_y \Delta \dot{\omega}_y - (I_z - I_x) \Delta \omega_x (\bar{\omega}_z + \Delta \omega_z) \\ &= I_y \Delta \dot{\omega}_y - (I_z - I_x) \Delta \omega_x \bar{\omega}_z - (I_z - I_x) \Delta \omega_x \Delta \omega_z \\ &= I_y \Delta \dot{\omega}_y - (I_z - I_x) \bar{\omega}_z \Delta \omega_x. \end{aligned}$$

From $I_z \dot{\omega}_z = (I_x - I_y) \omega_x \omega_y + M_z$,

$$\begin{aligned} \Rightarrow M_z &= I_z \dot{\omega}_z - (I_x - I_y) \omega_x \omega_y \\ &= I_z (\bar{\omega}_z^0 + \Delta \dot{\omega}_z) - (I_x - I_y) \Delta \omega_x \Delta \omega_y \\ &= I_z \Delta \dot{\omega}_z. \end{aligned}$$

$$\therefore \begin{cases} I_x \Delta \dot{\omega}_x - (I_y - I_z) \bar{\omega}_z \Delta \omega_y = M_x \\ I_y \Delta \dot{\omega}_y - (I_z - I_x) \bar{\omega}_z \Delta \omega_x = M_y \\ I_z \Delta \dot{\omega}_z = M_z. \end{cases} \rightarrow \text{Coupled}$$

$I_z \Delta \dot{\omega}_z = M_z$ is decoupled.

$$\begin{cases} \Delta \dot{\omega}_x = \frac{I_y - I_z}{I_x} \bar{\omega}_z \Delta \omega_y + \frac{M_x}{I_x} \\ \Delta \dot{\omega}_y = \frac{I_z - I_x}{I_y} \bar{\omega}_z \Delta \omega_x + \frac{M_y}{I_y} \\ \Delta \dot{\omega}_z = \frac{M_z}{I_z} \end{cases}$$

$$\Rightarrow \begin{bmatrix} \Delta \dot{\omega}_x \\ \Delta \dot{\omega}_y \\ \Delta \dot{\omega}_z \end{bmatrix} = \begin{bmatrix} 0 & \frac{I_y - I_z}{I_x} \bar{\omega}_z & 0 \\ \frac{I_z - I_x}{I_y} \bar{\omega}_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta \omega_x \\ \Delta \omega_y \\ \Delta \omega_z \end{bmatrix} + \begin{bmatrix} M_x/I_x \\ M_y/I_y \\ M_z/I_z \end{bmatrix}$$

$$\begin{vmatrix} \lambda & -\frac{I_y - I_z}{I_x} \bar{\omega}_z & 0 \\ -\frac{I_z - I_x}{I_y} \bar{\omega}_z & \lambda & 0 \\ 0 & 0 & \lambda \end{vmatrix} = (-1)^{3+3} \lambda \begin{vmatrix} \lambda & -\frac{I_y - I_z}{I_x} \bar{\omega}_z \\ -\frac{I_z - I_x}{I_y} \bar{\omega}_z & \lambda \end{vmatrix}$$

By Cramer's rule.

$$= \lambda \left(\lambda^2 + \frac{(I_y - I_z)(I_x - I_z)}{I_x I_y} \bar{\omega}_z^2 \right) = 0.$$

↳ Characteristic equation.

① If I_z is major ($I_z > I_y$ & $I_z > I_x$) OR minor ($I_z < I_y$ & $I_z < I_x$), then eigenvalues are

$$\lambda = 0, \pm j \bar{\omega}_z \sqrt{\frac{(I_y - I_z)(I_x - I_z)}{I_x I_y}}$$

Hence the system is marginaly stable because there exist one zero and two pure imaginary.

② If I_z is intermediate, ($I_x < I_z < I_y$ OR $I_y < I_z < I_x$) then eigenvalues are

$$\lambda = 0, \pm \bar{\omega}_z \sqrt{\frac{(I_y - I_z)(I_z - I_x)}{I_x I_y}}$$

Thus the system is unstable because there exists a positive real eigenvalue.

③ If $I_z = I_x$ OR $I_z = I_y$, then all eigenvalues are zero.

$$\lambda = 0, 0, 0.$$

Then the system is unstable because there are many zero eigenvalues.

($I_x = I_y \neq I_z$ 인 경우는 ①과 같고, $I_x = I_y = I_z$ 인 경우는 ③과 같음)

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$$I = \text{diag}(200, 200, 200) \text{ kg m}^2 \quad \vec{a} = \begin{bmatrix} 0.424 \\ -0.1566 \\ 0.1707 \end{bmatrix}$$

$$\text{rest to rest} \Rightarrow \dot{\phi}_0 = 0 = \dot{\phi}_f$$

$$R_0 = \exp[\xi(0)] \Rightarrow \phi_0 = 1$$

$$R_f = I = \exp[0] \Rightarrow \phi_f = 0$$

$$\text{Assume } \ddot{\phi}(t) = \begin{cases} -0.01 \text{ rad/sec} & \text{for } 0 \leq t \leq t_s \\ 0.01 \text{ rad/sec} & \text{for } t_s < t \leq t_f \end{cases}$$

$$\text{From } R = \exp(s(\vec{a})\phi), \quad \dot{R} = s(\vec{a})\dot{\phi}(t) \exp(s(\vec{a})\phi) = s(\vec{a})\dot{\phi}(t) R$$

$$\text{Since } \dot{R} = S(\vec{\omega})R \quad \text{where } S(\cdot): \mathbb{R}^3 \rightarrow \mathbb{R}^{3 \times 3} \text{ is skew-symmetric mapping, then } S(\vec{\omega})R = \dot{R} = s(\vec{a})\dot{\phi}(t)R = S(\vec{a}\dot{\phi})R$$

$$\therefore \vec{\omega} = \vec{a}\dot{\phi}$$

$$\text{If } 0 \leq t \leq t_s,$$

$$\dot{\phi}(t) = \int_0^t \ddot{\phi}(\tau) d\tau + \dot{\phi}(0)$$

$$= \int_0^t -0.01 d\tau + 0$$

$$= -0.01t$$

$$\text{Thus } \dot{\phi}(t_s) = -0.01t_s$$

$$\text{If } t_s < t \leq t_f,$$

$$\dot{\phi}(t) = \int_0^t \ddot{\phi}(\tau) d\tau + \dot{\phi}(0)$$

$$= \int_0^{t_s} \ddot{\phi}(\tau) d\tau + \int_{t_s}^t \ddot{\phi}(\tau) d\tau + \dot{\phi}(0)$$

$$= -0.01t_s + \int_{t_s}^t 0.01 d\tau + 0$$

$$= -0.01t_s + 0.01(t - t_s) = 0.01(t - 2t_s)$$

$$\therefore \vec{\omega} = \vec{a}\dot{\phi}(t) = \begin{cases} -0.01t\vec{a} & \text{for } 0 \leq t \leq t_s \\ 0.01(t - 2t_s)\vec{a} & \text{for } t_s < t \leq t_f \end{cases}$$

$$\vec{\omega}(0) = 0 = \vec{\omega}(t_f) \quad (\because \text{rest to rest maneuver})$$

From Euler's rotational equations,

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$$\begin{aligned}\vec{M} &= I\dot{\vec{\omega}} + \vec{\omega} \times I\vec{\omega} \\ &= I\ddot{\phi}(t)\vec{a} + \dot{\phi}(t)\vec{a} \times I\dot{\phi}(t)\vec{a} \\ &= 200\ddot{\phi}(t)\vec{a} + 200\dot{\phi}(t)\vec{a} \times \dot{\phi}(t)\vec{a} = 200\ddot{\phi}(t)\vec{a}\end{aligned}$$

Hence control moment vector is,

$$\vec{M} = \begin{cases} -2\vec{a} & \text{for } 0 \leq t \leq t_s \\ 2\vec{a} & \text{for } t_s < t \leq t_f \end{cases}$$

By the boundary condition, $\vec{\omega}(0) = 0 = \vec{\omega}(t_f)$.

$$\vec{\omega}(0) = -0.01 \times 0 \times \vec{a} = 0.$$

$$\vec{\omega}(t_f) = 0.01(t_f - 2t_s)\vec{a} = 0.$$

$$\text{Since } \vec{a} \neq 0, \quad t_f - 2t_s = 0. \quad \therefore t_s = \frac{t_f}{2}$$

If $0 \leq t \leq t_s = t_f/2$,

$$\begin{aligned}\phi(t) &= \int_0^t \dot{\phi}(\tau) d\tau + \phi(0) \\ &= \int_0^t (-0.01\tau) d\tau + 1 \\ &= -0.005t^2 + 1\end{aligned}$$

$$\text{Then } \phi(t_s) = \phi(t_f/2) = -0.005\left(\frac{t_f}{2}\right)^2 + 1 = -\frac{1}{800}t_f^2 + 1.$$

If $t_s = t_f/2 < t \leq t_f$,

$$\begin{aligned}\phi(t) &= \int_0^t \dot{\phi}(\tau) d\tau + \phi(0) \\ &= \int_0^{t_f/2} (-0.01\tau) d\tau + \int_{t_f/2}^t 0.01(\tau - t_f) d\tau + 1 \\ &= -\frac{1}{800}t_f^2 + \int_{t_f/2}^t 0.01\tau d\tau - \int_{t_f/2}^t 0.01t_f d\tau + 1 \\ &= -\frac{1}{800}t_f^2 + \frac{1}{200}\left(t^2 - \frac{t_f^2}{4}\right) - \frac{1}{100}t_f\left(t - \frac{t_f}{2}\right) + 1\end{aligned}$$

$$\begin{aligned}\text{Then } \phi(t_f) &= -\frac{1}{800}t_f^2 + \frac{1}{200}\left(t_f^2 - \frac{t_f^2}{4}\right) - \frac{1}{100}t_f\left(t_f - \frac{t_f}{2}\right) + 1 \\ &= -\frac{1}{800}t_f^2 + \frac{t_f^2}{200} \cdot \frac{3}{4} - \frac{t_f^2}{100} \cdot \frac{1}{2} + 1\end{aligned}$$

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$$= -\frac{1}{800}t_f^2 + \frac{3}{800}t_f^2 - \frac{1}{200}t_f^2 + 1$$

$$= \frac{-1+3-4}{800}t_f^2 + 1$$

$$= -\frac{1}{400}t_f^2 + 1$$

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since $\phi(t_f) = 0$, $-\frac{1}{400}t_f^2 + 1 = 0 \Rightarrow t_f^2 = 400$.

since $t_f > 0 \Rightarrow t_f = 20 \text{ sec}$

$$I_y \ddot{\Delta\theta} = 3\omega_0^2 (-I_x + I_z) \Delta\theta + M_y$$

$$\Rightarrow 200 \ddot{\Delta\theta} = 3\omega_0^2 (-150 + 100) \Delta\theta + M_y$$

$$\Rightarrow \ddot{\Delta\theta} = -\frac{3}{4}\omega_0^2 \Delta\theta + \frac{M_y}{200}$$

$$\Rightarrow \begin{bmatrix} \dot{\Delta\theta} \\ \ddot{\Delta\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{3}{4}\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta\theta \\ \dot{\Delta\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{M_y}{200}$$

$\Delta\theta_0 = 45^\circ = \pi/4 \text{ rad}$. $\Delta\theta_f = 0^\circ = 0 \text{ rad}$.

rest to rest $\Rightarrow \dot{\Delta\theta}_0 = 0 = \dot{\Delta\theta}_f$

Let assume that t_s is the time that the single impulse is applied.

Then if $0 \leq t \leq t_s$, the spacecraft move without external moment.

$$\Rightarrow \begin{bmatrix} \dot{\Delta\theta}(t) \\ \ddot{\Delta\theta}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \sqrt{\frac{4}{3}}\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta\theta(t) \\ \dot{\Delta\theta}(t) \end{bmatrix}$$

Let $a^2 = \frac{4}{3}\omega_0^2$; $a > 0$. ($a = \frac{2\sqrt{3}}{3}\omega_0 \approx 1.3856 \times 10^{-3} \text{ rad/sec}$)

$$\begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix} = \begin{bmatrix} -a^2 & 0 \\ 0 & -a^2 \end{bmatrix} = -a^2 \mathbf{I}$$

$$\Rightarrow \begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix}^3 = -a^2 \begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix}^4 = -a^2 \begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix}^2 = (-a^2)^2 \mathbf{I}$$

$$\Rightarrow \exp\left(\begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix} t\right) = \mathbf{I} + \begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix} t + \frac{1}{2!} \begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix}^2 t^2 + \dots$$

$$= \mathbf{I} \left(1 + \frac{1}{2!} (-a^2) t^2 + \frac{1}{4!} (-a^2)^2 t^4 + \dots\right)$$

$$+ \begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix} \left(t + \frac{1}{3!} (-a^2) t^3 + \frac{1}{5!} (-a^2)^2 t^5 + \dots\right)$$

0324.032 이진아.

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$$= I \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} (at)^{2k} \right) + \frac{1}{a} \begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix} \left(\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(2k-1)!} (at)^{2k-1} \right)$$

$$= I \cos(at) + \begin{bmatrix} 0 & a^{-1} \\ -a & 0 \end{bmatrix} \sin(at)$$

$$= \begin{bmatrix} \cos at & \sin at / a \\ -a \sin at & \cos at \end{bmatrix}$$

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$$\Rightarrow \begin{bmatrix} \Delta \theta(t) \\ \Delta \dot{\theta}(t) \end{bmatrix} = \exp \left(\begin{bmatrix} 0 & 1 \\ -a^2 & 0 \end{bmatrix} t \right) \begin{bmatrix} \Delta \theta(0) \\ \Delta \dot{\theta}(0) \end{bmatrix}$$

$$= \begin{bmatrix} \cos at & \sin at / a \\ -a \sin at & \cos at \end{bmatrix} \begin{bmatrix} \pi/4 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} \Delta \theta(t_s) = \frac{\pi}{4} \cos at_s \\ \Delta \dot{\theta}(t_s) = -\frac{\pi}{4} a \sin at_s \end{cases}$$

Change in the pitch rate due to impulse

최종을 준 직후, When $t = t_s^+$

$$\Delta \theta(t_s^+) = \frac{\pi}{4} \cos at_s \quad \& \quad \Delta \dot{\theta}(t_s^+) = -\frac{\pi}{4} a \sin at_s + \Delta \dot{\theta}_s$$

최종 시간 t_s 에서의 상태는,

$$\begin{bmatrix} \Delta \theta(t_f) \\ \Delta \dot{\theta}(t_f) \end{bmatrix} = \begin{bmatrix} \cos a(t_f - t_s) & \sin a(t_f - t_s) / a \\ -a \sin a(t_f - t_s) & \cos a(t_f - t_s) \end{bmatrix} \begin{bmatrix} \frac{\pi}{4} \cos at_s \\ -\frac{\pi}{4} a \sin at_s + \Delta \dot{\theta}_s \end{bmatrix}$$

↓
STM

STM의 determinant를 구해보면 $\cos^2 a(t_f - t_s) + \sin^2 a(t_f - t_s) = 1$

이때 $\ker(STM) = \{ \vec{0} \}$ 이다.

따라서 최종 상태 ($\Delta \theta_f = 0$ & $\Delta \dot{\theta}_f = 0$)를 만족하는 $\Delta \theta(t_s^+)$, $\Delta \dot{\theta}(t_s^+)$ 가 유일하게 결정된다.
 → 따라서 주어진 마지막 순간에 주어야 한다. ($t_s = t_f$).
 (정리 마지막)

$$\Delta \theta_f = 0 \quad \& \quad \Delta \dot{\theta}_f = 0 \quad \text{이때} \quad \Delta \theta(t_s^+) = 0 \quad \& \quad \Delta \dot{\theta}(t_s^+) = 0 \quad \text{이어야 한다?}$$

$$\Rightarrow \frac{\pi}{4} \cos at_s = 0 \quad \& \quad -\frac{\pi}{4} a \sin at_s + \Delta \dot{\theta}_s = 0$$

$$\Rightarrow at_s = \left(\frac{2n-1}{2} \right) \pi \quad \text{for } n \in \mathbb{Z}^+ \cup \{0\}$$

$$\Rightarrow t_s = 1133.6246 + 226\pi.2492n_{sec} \quad \text{for } n \in \mathbb{Z}^+ \cup \{0\}$$

If n is even OR 0, then $\Delta \dot{\theta}_s = \frac{\pi}{4} a = 1.0883e-03 \text{ rad/sec}$

If n is odd, then $\Delta \dot{\theta}_s = -\frac{\pi}{4} a = -1.0883e-03 \text{ rad/sec}$

Change in the pitch rate due to impulse

Thus ~~324032~~ ³²⁴⁰³² ~~이전~~ ^{이전} to rest maneuver can be performed using open $\pi/5$
 a single impulsive if given transfer time is greater than 1133.6246 sec.
 The impulse should be performed at t_s , and it is same as transfer time.

이전 시간을 설정하는 경우 특정 시간 $(1133.6246(2n-1) \text{ } n \in \mathbb{Z}^+)$ 동안만 위상이 이동한다.

$\Rightarrow$ Change in the pitch rate due to impulse $\frac{10}{10}$.

$$\Delta \theta_s = \begin{cases} 1.0883 \times 10^{-3} \text{ rad/sec} & \text{if } n \equiv 0 \pmod{2} \\ -1.0883 \times 10^{-3} \text{ rad/sec} & \text{if } n \equiv 1 \pmod{2} \end{cases}$$

where $n \in \mathbb{Z}^+ \cup \{0\}$.

~~Transfer time impulse rate~~